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Effective elastic properties of cracked solids: critical review of some basic concepts

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The problem of effective moduli of cracked solids is critically reviewed. Various approaches to the problem are discussed; they are further assessed by comparing their predictions to results for sample deterministic arrays. These computer experiments indicate that *the approximation of non-interacting cracks has a wider than expected range of applicability.*

Some of the deficiencies of various approximate schemes seem to be related to inadequacy of the conventionally used *crack density parameter* (insensitive to mutual positions of cracks). An alternative parameter that *has* this sensitivity, is suggested.

Finally, the problem of effective moduli is discussed in the context of "damage mechanics". It is argued that, contrary to the spirit of many damage models, there is *no direct quantitative correlation* between progression of a microcracking solid towards fracture and deterioration of its stiffness; thus, the effective moduli may not always serve as a reliable indicator of damage.

1. INTRODUCTION

The problem of effective elastic properties of cracked solids is of considerable interest for solid mechanics, geophysics and material science; it is particularly important for constitutive modelling of brittle microcracking materials.

This review focusses on several key issues of the problem; their choice reflects, to some extent, interests of the author.

Solid with cracks vs solid with inclusions. The problem of effective elastic moduli of cracked solids can formally be regarded as a limiting case of the more general problem for a matrix containing inclusions: in this limiting case, (a) inclusions shrink to surfaces and (b) their elastic moduli tend to zero.

Such an approach leads to some difficulties, however:

(1) The results may depend on the *order* of the mentioned transitions to limit. The proper one first sets the inclusion's moduli to be zero (thus creating a cavity) and then shrinks the cavity to a crack. An attempt to reverse the order leads to a difficulty: in the framework of linear elasticity, infinitesimally thin platelets of finite stiffness do not affect the effective moduli, thus becoming "invisible", and the second transition to limit cannot be made (unless it is combined with the first one and a special constraint is imposed).

The commonly used crack density parameter (introduced by Bristow, 1960 and Walsh, 1965) is (upper indices will indicate crack numbers in the following):

$$\rho = \frac{1}{A}\sum l^{(i)2} \qquad \text{or} \qquad \rho = \frac{1}{V}\sum l^{(i)3} \tag{1.1}$$

in *2-D* (rectilinear cracks of lengths $2l^i$, A is the representative area) and *3-D* (circular cracks of radii l^i, V is the representative volume element, RVE), respectively. (Some definitions include a multiplier π in the *2-D* case, then ρ is the relative area of circles with cracks as diameters). For cracks of *non-circular* shapes in *3-D*, ρ was generalized by Budiansky and O'Connell (1976) to

$$\rho = \frac{2}{\pi}\frac{1}{V}\sum (S^2/P)^{(i)} \tag{1.1a}$$

where S and P are the crack area and perimeter.

(3) Some approaches of mechanics of composite materials *degenerate* for cracked solids. *Bounding* of the effective moduli is an example: neither upper nor lower non-trivial bounds (that would apply to *any* sample of a given crack statistics) can be established for cracked solids. Another example is Mori-Tanaka's method for materials with interacting inclusions: being applied to cracked solids, it yields results corresponding to *non-interacting* cracks.

A direct approach to the problem of effective moduli of cracked solids (rather than the one based on mechanics of two components materials) appears, therefore, more appropriate.

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Average strains and stresses in a cracked solid. Since displacements are, generally, discontinuous across crack surfaces S^i, the strains are singular at S^i. Thus, the strain field in a solid with cracks is a sum of regular and singular parts:

$$\varepsilon(\mathbf{x}) = \mathbf{M} : \sigma(\mathbf{x}) + \frac{1}{2}\sum_i (\mathbf{bn} + \mathbf{nb})^i \, \delta(S^i) \qquad (1.2)$$

where, in representing the regular part as $\mathbf{M}:\sigma$, the matrix material is assumed to be linear elastic (with compliance tensor $\mathbf{M}$) and $\delta(S^i)$ is a delta function concentrated on S^i (as a derivative of a unit step, it has the dimension of length^{-1}). A colon denotes contraction over two indices; $\mathbf{bn}$, $\mathbf{nb}$ denote dyadic (tensor) products of the displacement discontinuity vector $\mathbf{b} = \mathbf{u}^+ - \mathbf{u}^-$ (crack opening displacement, COD) and unit normal $\mathbf{n}$ to a crack. Vectors $\mathbf{b}$ and $\mathbf{n}$ are, generally, variable along cracks; their directions are coordinated by defining the "+" side of S as corresponding to the positive sense of $\mathbf{n}$ (then change of sign of $\mathbf{n}$ changes the sign of $\mathbf{b}$ and the products $\mathbf{bn}$, $\mathbf{nb}$ remain invariant).

In the following, we assume the matrix material to be elastically homogeneous: $\mathbf{M}(\mathbf{x}) = const = \mathbf{M}^0$.

Averaging (1.2) over V and using the property of $\delta(S)$ that $\int_V f(x)\delta(S)dV$ reduces to the surface integral $\int_S f(x)dS$ yields the volume average of strain:

$$\langle\varepsilon\rangle = \mathbf{M}^0 : \langle\sigma\rangle + \frac{1}{2V}\sum_i \int_{S_i} (\mathbf{bn}+\mathbf{nb})dS$$
$$= (\mathbf{M}^0+\Delta\mathbf{M}) : \langle\sigma\rangle \equiv \mathbf{M} : \langle\sigma\rangle \qquad (1.3)$$

where $\Delta\mathbf{M}$ is the change in compliance due to cracks and $\mathbf{M}$ is the effective compliance. Linear dependence of $\mathbf{b}^i$ on $\langle\sigma\rangle$ (used in the second line of 1.3) assumes linear elasticity of the matrix and absence of friction on cracks (if friction *is* present, then 1.3 is to be rewritten in the incremental form; $\Delta\mathbf{M}$ will depend on $\langle\sigma\rangle$ and on the direction of $\langle d\sigma\rangle$).

Volume averages $\langle\varepsilon\rangle$ and $\langle\sigma\rangle$ can be expressed in terms of quantities defined on the outer boundary Γ of V, i.e., in terms of experimentally accessible (measurable) quantities. Since, as follows from the divergence theorem, $\langle\varepsilon\rangle$ for a simply connected region is given by the integral $(1/2V)\int_\Gamma (\mathbf{un}+\mathbf{nu})d\Gamma$, this integral *defines* $\langle\varepsilon\rangle$ for a volume element of the *effective* (homogenized) material. Thus, if *displacements* are prescribed at the boundary of V, then, by definition, *introduction of cracks into V does not change* $\langle\varepsilon\rangle$. At the microstructural level, it produces redistribution of strains: due to the contribution from displacement discontinuities into the constant total $\langle\varepsilon\rangle$, strains in the matrix decrease.

Average stresses are defined in terms of quantities on Γ as $(1/V)\int_\Gamma \mathbf{xT}d\Gamma$, where $\mathbf{x}$ is a position vector (counted from the centroidal point) and $\mathbf{T}$ is a traction vector. This definition is justified by the fact that, as follows from the divergence theorem, $\langle\sigma\rangle = (1/V)\int_\Gamma \mathbf{xT}d\Gamma$, for *both* simply connected regions and regions with traction-free cracks, provided the body force is constant in V (as noted by Lehner,1992, the latter requirement can be relaxed). Thus, by definition, *introduction of cracks does not change the average stress*, if tractions are prescribed on the boundary. This means that, if a certain stress component is locally amplified by a crack (say, has a singularity near the tip), this is compensated by "unloading" at other points of a solid.

If stress is constant ($\equiv\sigma^0$) along Γ ("homogeneous boundary conditions", Hashin,1983), then another application of the divergence theorem yields $\sigma^0 = \langle\sigma\rangle$.

If the RVE constitutes a part of a statistically homogeneous field of cracks and is sufficiently large to be representative, then it can be reasonably hypothesized that substitution of the actual stress on Γ (generally, fluctuating along Γ) by its constant average will affect only a "boundary layer" of V (having thickness of the order of the fluctuation wavelength) and will not significantly change $\langle\sigma\rangle$, so that $\sigma^0 \approx \langle\sigma\rangle$. If, in addition, the remotely applied σ^∞ is such that, in the absence of cracks the stress field would have been homogeneous, then σ^0 can be approximately identified with σ^∞ (for a detailed discussion of statistical homogeneity and related issues, see Hashin,1983 and Beran,1968). We add that the identification $\sigma^0 = \langle\sigma\rangle = \sigma^\infty$ is *exact* for non-interacting cracks, when cracks inside the RVE experience no influence of those outside.

In the following, we omit the averaging signs $\langle\ \rangle$ for stresses and strains, assuming $\varepsilon \equiv \langle\varepsilon\rangle$ and $\sigma \equiv \langle\sigma\rangle = \sigma^\infty$.

For *flat* cracks, $\mathbf{n} = const$ within each crack; then

$$\varepsilon = \mathbf{M}^0 : \sigma + \frac{1}{2V}\sum_i (\langle\mathbf{b}\rangle\mathbf{n}+\mathbf{n}\langle\mathbf{b}\rangle)^i S^i \qquad (1.4)$$

(volume V and areas S^i to be changed to area A and lengths $2l^i$ in the 2-D case) where $\langle\mathbf{b}^i\rangle$ is $\mathbf{b}^i$ averaged over S^i.

Thus, contribution of a given crack into the overall strain is proportional to the product (crack area)$\langle\mathbf{b}\rangle$. The unknowns are vectors $\langle\mathbf{b}^i\rangle$ as functions of σ.

Representations of the type of (1.3) or (1.4) were first given, probably, by Hill (1963) for a more general case of arbitrary cavities; he expressed the cavity's contribution into the overall strain as an integral over the cavity surface

$$(-1/2V_s)\int_S (\mathbf{un}+\mathbf{nu})dS$$

where V_s is the volume of the solid phase; it reduces to the second term of (1.3) when cavities shrink to cracks.

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Such representations are routinely used as a starting point for analysis (Kachanov,1980a,b,1982, Hashin,1983, and many other authors; Horii and Nemat-Nasser,1983 remarked, additionally, that $\Delta\mathbf{M}$ can be expressed in terms of the Green's function for a solid with cracks; as they noted, however, this is not very useful since such functions are difficult to evaluate).

Elastic potential (complementary energy density) of a cracked solid. Elastic potential f of the effective material is obtained by contracting $\sigma_{ij}/2$ with ε_{ij}; taking the latter from (1.4) yields f in terms of "microstructural" quantities defined on S^i:

$$f(\sigma) = \frac{1}{2}\sigma:\mathbf{M}^{eff}:\sigma = \frac{1}{2}\sigma:\mathbf{M}^{o}:\sigma + \frac{1}{2V}\sum_i (\mathbf{n}\cdot\sigma\cdot\langle\mathbf{b}\rangle)^i S^i$$
$$\equiv f_o(\sigma) + \Delta f \qquad (1.5)$$

(to be appropriately modified in the *2-D* case) where $f_o(\sigma)$ is the potential of a matrix without cracks. For an isotropic matrix with Young's modulus E_o and Poisson's ratio ν_o

$$f_o(\sigma) = \frac{1+\nu_o}{2E_o}\sigma_{ij}\sigma_{ij} - \frac{\nu_o}{2E_o}(\sigma_{kk})^2 \qquad (1.6)$$

where, in the *2-D* case, E_o changes to E_o' ($=E_o$ for plane stress and $=E_o/(1-\nu_o^2)$ for plane strain).

Expression (1.5) plays a central role in further analyses.

Crack compliance tensor. We introduce tensor **B** that interrelates a vector of uniform traction $\boldsymbol{t}$ applied at the crack faces and the vector of resulting average COD:

$$\langle\mathbf{b}\rangle = \mathbf{t}\cdot\mathbf{B} \qquad (1.7)$$

The second rank tensor **B** can be called *crack compliance tensor*. It depends on crack size and shape, on elastic properties of the matrix and, in the case of anisotropic matrix, on orientation of the crack with respect to the anisotropy axes of the matrix. In the case of a body of finite size, it also depends on body's geometry.

In the case of an isolated crack in an infinite body with stress σ at infinity (such that, in the absence of a crack the stress state is uniform), (1.7) can be rewritten as

$$\langle\mathbf{b}\rangle = \mathbf{n}\cdot\sigma\cdot\mathbf{B} \qquad (1.8)$$

In the coordinate system (**n, t**) where **t** is tangential to the crack, the diagonal components B_{nn}, B_{tt} characterize the normal (shear) COD's produced by the normal (shear) tractions. The off-diagonal components characterize *coupling of the modes* (relevant for anisotropic matrix, and for non-circular cracks in isotropic matrix). Betti's reciprocity theorem implies that $B_{tn} = B_{nt}$ i.e. **B** is symmetric.

Compliance tensor of a crack in an infinite isotropic solid. In the case of *isotropic* matrix, normal and shear modes are *uncoupled.*

In *2-D*, **B** is proportional to a unit tensor:

$$\mathbf{B} = (\pi l/E_o')\mathbf{I} \qquad (1.8)$$

reflecting equal compliances in normal and shear modes.

In *3-D*, for a *circular* crack (radius l), $\mathbf{B} = \beta\,\mathbf{nn} + \gamma(\mathbf{tt}+\mathbf{ss})$ where $\boldsymbol{s}$ and $\boldsymbol{t}$ are any two orthogonal vectors in the crack plane, and $\beta = 16l(1-\nu_o^2)/3\pi E_o$; $\gamma = \beta/(1-\nu_o/2)$. Note that **B** is not proportional to **I**, since normal and shear compliances of a crack are different. Since $\mathbf{tt}+\mathbf{ss}+\mathbf{nn} = \mathbf{I}$, we can rewrite B in the form

$$\mathbf{B} = \gamma\left(\mathbf{I} - \frac{\nu_o}{2}\mathbf{nn}\right) \qquad (1.9)$$

The ***I***-term represents the part of COD that is *collinear* to the traction $\mathbf{n}\cdot\sigma$; the ***nn***-term characterizes the deviation from collinearity. An important observation is that the coefficient at the ***nn***-term is much smaller than the one at the ***I***-term, reflecting a relatively small difference in normal and shear compliances of a circular crack (vanishing at $\nu_o=0$).

For an *elliptical* crack, **B** can be constructed by utilizing Eshelby's results on ellipsoidal inclusions (specialized for elliptical cracks by Budiansky and O'Connell, 1976):

$$\mathbf{B} = \xi\mathbf{nn} + \eta\mathbf{ss} + \zeta\mathbf{tt} \qquad (1.10)$$

where $\boldsymbol{s}$ and $\boldsymbol{t}$ are unit vectors of the major, $2a$, and minor, $2b$, ellipse's axes. Coefficients ξ, η, ζ are expressed in terms of elliptic integrals: $\xi = 4(1-\nu_o^2)b/(3E_oE(k))$ where $E(k)$ is the complete elliptic integral of the first kind of the argument $k = (1-(b/a)^2)^{1/2}$ and η, ζ are obtained from ξ by changing $E(k)$ to Q^{-1} and R^{-1}, respectively, with Q and R given by formula (31) of Budiansky and O'Connell. Using the identity $\mathbf{nn}+\mathbf{ss}+\mathbf{tt} = \mathbf{I}$, we transform (1.10) to the form

$$\mathbf{B} = \frac{\eta+\zeta}{2}\left[\mathbf{I} + \left(\frac{2\xi}{\eta+\zeta} - 1\right)\mathbf{nn} + \frac{\eta-\zeta}{\eta+\zeta}(\mathbf{ss}-\mathbf{tt})\right] \qquad (1.11)$$

where the first two terms are similar to **B** of a circular crack, and the third term emerges due to deviation of the crack shape from circular.

B-tensors are particularly useful in the cases of non-circular cracks and cracks in anisotropic matrix.

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2. NON-INTERACTING CRACKS. CRACK DENSITY TENSOR. ORTHOTROPY.

For non-interacting cracks (in the *isotropic* matrix material), the effective moduli can be found exactly, for an arbitrary crack orientation statistics, in both *2-D* and *3-D*. It is the simplest (and the *only* non-controversial) approximation to the problem; at the same time, it has a wider than expected range of applicability (see sections 7,9).

It is often called the approximation of small crack density (in fact, the two names are usually used as synonyms). We refrain, however, from using this term, since *these two assumptions are generally not equivalent.* Indeed, if crack locations are non-random, interactions can be strong (and the discussed approximation inapplicable) at vanishingly small density (section 10). On the other hand, in the case of *random* crack locations, the approximation of non-interacting cracks produces accurate results at high crack densities, due to cancellation of the competing effects of stress shielding and amplification (section 9).

In the approximation of non-interacting cracks, each crack is regarded as an *isolated* one; it is embedded into the σ-field and does not experience any influence of other cracks. Then $\langle \mathbf{b} \rangle$ for each crack is determined by $\mathbf{n}\cdot\sigma$. Mutual positions of cracks do not matter in this approximation, hence averaging over a crack array is reduced to summation over orientations. Since the change in compliance ΔM due to cracks is a sum of the isolated cracks' contributions, *the compliance is linear in crack density parameter.*

Such calculations were first done for randomly oriented cracks, in *2-D* and *3-D*, by Bristow (1960). They were repeated and extended to the case of frictional contact between crack faces and to the case of spherical cavities by Walsh (1965), who was apparently unaware of Bristow's results.

Since averaging is reduced simply to integration over orientations, such calculations are easy to repeat for *an arbitrary* (non-random) *orientational distribution.* Such calculations were done by a number of authors. For one or several families of parallel cracks, they were explicitly written down by Piau (1980). They are implicitly (in the form of elastic potential) given, for arbitrary orientation distributions, by Kachanov (1980a); they are also contained, explicitly or implicitly, in papers on self-consistent and differential schemes (see, for example, Hoenig, 1979).

The results can be written in the most general form (arbitrary orientational distribution and crack shapes) by introducing (1.8) into (1.5). This yields the elastic potential:

$$\Delta f = (1/2)\,\sigma :: \Delta M :: \sigma \qquad (2.1)$$

where

$$\Delta M = (1/A)\sum l^i\,(\mathbf{nBn})^i \quad \text{in } 2\text{-}D$$

$$= (1/2V)\sum S^i\,(\mathbf{nBn})^i \quad \text{in } 3\text{-}D \qquad (2.2)$$

is the additional compliance due to cracks (ΔM_{ijkl} is to be appropriately symmetrized, with respect to $i\leftrightarrow j, k\leftrightarrow l, ij\leftrightarrow kl$).

The analysis below is based on the concept of *crack density tensor* (introduced by Vakulenko and Kachanov,1971, Kachanov,1972 and, in a more complete and corrected form, by Kachanov,1980a) that emerges naturally as a term in the elastic potential. The effective moduli for arbitrary crack statistics are then obtained in a simple unified way, avoiding averaging over orientations (instead, one just has to solve the eigenvalue problem for a second rank tensor). This concept also clarifies symmetry of the effective properties.

Two-dimensional case. Vectors $\langle \mathbf{b}^i \rangle$ and $\mathbf{n}^i\cdot\sigma$ are *collinear*:

$$\langle \mathbf{b}^i \rangle = (\pi l^i / E_o')\,\mathbf{n}^i\cdot\sigma \qquad (2.3)$$

since *"compliances" of a crack are the same in shear and normal modes.* This collinearity has important implications for symmetry of the effective properties, as seen below.

Substituting (2.3) into potential (1.5) (or, equivalently, **B**- tensor given by (1.8) into 2.1,2), one obtains

$$\Delta f = (\pi / E_o')\,(\sigma\cdot\sigma) :: \alpha \quad = (\pi / E_o')\sigma_{ij}\sigma_{jk}\alpha_{ik} \qquad (2.4)$$

where

$$\alpha = \frac{1}{A}\sum_i (l^2 \mathbf{n}\,\mathbf{n})^i \qquad (2.5)$$

is a second rank *crack density tensor* (summation may be substituted by integration over crack orientations, if computationally convenient).

Since $\mathbf{n}\cdot\mathbf{n}=1$, the linear invariant $tr\,\alpha = \alpha_{kk}$ is the conventional scalar crack density:

$$r\,\alpha = \frac{1}{A}\sum_i l^{(i)2} = \rho \qquad (2.6)$$

so that α is a natural tensorial generalization of ρ that accounts for crack orientations.

Thus, elastic potential is a function of σ and α:

$$f(\sigma, \alpha) = f_0(\sigma) + \Delta f(\sigma, \alpha) \qquad (2.7)$$

where Δf is a *simultaneous invariant* of σ and α, quadratic in σ (linear elasticity at fixed α) and linear in α (non-interacting cracks). This reflects *isotropy* of the matrix: if both σ and α undergo the same orthogonal transformation (say, rotation) then the potential is insensitive to *any* such transformation.

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Equivalently to (2.4), we can specify the *compliance* increment due to cracks $\Delta M_{ijkl} = \partial^2 \Delta f / \partial\sigma_{ij}\partial\sigma_{kl}$ as

$$\Delta M_{ijkl} = (\pi / E_o') \, \delta_{\{ik}\alpha_{jl\}} \qquad (2.4a)$$

where the subscript braces denote all the appropriate symmetrizations (with respect to $i \leftrightarrow j$, $k \leftrightarrow l$, $ij \leftrightarrow kl$).

We emphasize that, for non-interacting cracks, these representations in terms of α are *rigorous*, and that they are a consequence of collinearity of vectors $\langle \mathbf{b}^i \rangle$ and $\mathbf{n}^i \cdot \sigma$.

Crack density tensor is *symmetric* (as a sum of symmetric dyads). Therefore, it has a principal representation

$$\alpha = \rho_1 \mathbf{e}_1\mathbf{e}_1 + \rho_2 \mathbf{e}_2\mathbf{e}_2 \qquad (2.8)$$

(ρ_1, ρ_2 and $\mathbf{e}_1, \mathbf{e}_2$ are principal values and unit vectors of α).

Each term in (2.8) represents a family of parallel cracks, therefore, *any system of cracks is fully equivalent, in its impact on the effective moduli, to two mutually perpendicular families of parallel cracks.*

As a consequence, the effective properties are *orthotropic* (rectangular symmetry) *for any statistics of crack orientations*; the orthotropy axes coincide with the principal axes of α. This result, *rigorous* for non-interacting cracks in *2-D*, may not be intuitively obvious (for example, it covers situations of the type of 2.17).

Moreover, the orthotropy is of a *simplified type*:

(1) since α is a second rank tensor with non-negative eigenvalues, the variation of elastic compliances with direction is described by *ellipses* (rather than complex curves of the fourth order, as in the case of general orthotropy);

(2) *the number of independent elastic constants is reduced from four* (general orthotropy in *2-D*) *to three*, due to the relation between shear and Young's moduli and Poisson's ratios (it follows from (2.4) and (2.8)):

$$1/G_{12} = (1+\nu_{12})/E_1 + (1+\nu_{21})/E_2 \qquad (2.9)$$

where x_1, x_2 are the principal axes of α. (Curiously, this relation was suggested in the theory of elasticity of anisotropic solids by Saint-Venant, as an approximation for reducing the number of constants, see Lekhnitsky, 1963);

(3) under uniaxial loading in one of the principal directions of α, *the presence of cracks does not cause any additional Poisson's effect*:

$$\nu_{12}/E_1 \ (=\nu_{21}/E_2) = \nu_o'/E_o' \qquad (2.10)$$

It is a consequence of α entering Δf through the invariant $(\sigma\cdot\sigma):\alpha$ only (although, in principle, one may construct one more invariant - $(\sigma:\alpha)tr\sigma$ - that is quadratic in σ and linear in α). It is also seen from the equivalence of an arbitrary crack array to two orthogonal families of parallel cracks: the considered uniaxial loading produces no CODs on one family and purely normal CODs on the other one.

This fact implies (in conjunction with 2.9) that the shear modulus can be expressed as

$$1/G_{12} = 1/E_1 + 1/E_2 + 2\nu_o/E_o' \qquad (2.9a)$$

Thus, *all elastic constants can be expressed in terms of two Young's moduli E_1, E_2 (in the principal directions of α) and one matrix constant ν_o/E_o'.*

The advantage of the representation (2.4) in terms of α is that the effective moduli are readily obtained for arbitrary orientational distribution of cracks *without averaging over orientations* as soon as this distribution is characterized by α. This is illustrated by the examples below.

A. *Isotropic (random) orientation distribution.* The crack density tensor is isotropic, and, since $tr\,\alpha = \rho$,

$$\alpha = (\rho/2)\mathbf{I} \qquad (2.11)$$

where $\mathbf{I}$ is a *2-D* unit tensor. Inserting (2.11) into (2.4) yields the elastic potential:

$$f(\sigma, \alpha) = f_o(\sigma) + (\pi/2E_o')\rho\, \sigma_{ij}\sigma_{ij} =$$

$$= \frac{1+\nu_o+\pi\rho}{2E_0'}\, \sigma_{ij}\sigma_{ij} - \frac{\nu_o}{2E_0'}(\sigma_{kk})^2 \qquad (2.12)$$

from which the effective moduli follow:

$$E' = E_o'(1+\pi\rho)^{-1}; \qquad \nu = \nu_o(1+\pi\rho)^{-1} \qquad (2.13)$$

B. *Parallel cracks* (normal to the x_1 axis). In this case

$$\alpha = \rho \mathbf{e}_1\mathbf{e}_1 \qquad (2.14)$$

and the potential takes the form

$$f(\sigma, \alpha) = f_o(\sigma) + (\pi/E_o')\rho\, \sigma_{1j}\sigma_{j1} \qquad (2.15)$$

from which one obtains E_1 and the shear modulus

$$E_1' = E_o'(1+2\pi\rho)^{-1}; \quad G_{12} = G_o[1+(2\pi G_o/E_o')\rho\,]^{-1} \qquad (2.16)$$

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Young's modulus in any other direction (inclined to cracks) is obtained by expressing α in the inclined coordinate system; variation of compliance with the angle of inclination is shown in fig. 2 (dashed line).

C. To illustrate convenience of the crack density tensor, we consider an example of *two families of parallel cracks inclined at an angle 30° to each other; one has the density twice as high as the other.* The crack density tensor is:

$$\alpha = (2\rho/3)\,\mathbf{n}^1\mathbf{n}^1 + (\rho/3)\,\mathbf{n}^2\mathbf{n}^2 \qquad (2.17)$$

where $\mathbf{n}^1$, $\mathbf{n}^2$ are unit normals to cracks of the first and second families. Solving the eigenvalue problem for α and substituting its principal representation into the potential, we find that the principal directions of orthotropy are rotated 9°.5 to $\mathbf{n}^1$ and that the Young's moduli in these directions are $E_o'/(1+5.86\rho)$ and $E_o'/(1+0.42\rho)$.

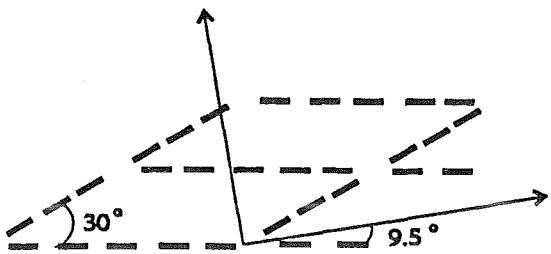


FIG.1. Orthotropy axes in the example C.

The case of several arbitrarily oriented families of parallel cracks is considered equally easily.

Three-dimensional case (circular cracks). Unlike the *2-D* case, $\langle \mathbf{b}^i \rangle$ is not collinear to $\mathbf{n}^i\cdot\sigma$, and the proportionality relation between $\langle \mathbf{b}^i \rangle$ and $\mathbf{n}^i\cdot\sigma$ should be written for the normal and shear components separately:

$$\langle \mathbf{b}^i \rangle = \beta \frac{l^i}{E_o}\mathbf{n}^i\cdot\sigma\cdot\mathbf{n}^i\mathbf{n}^i + \gamma \frac{l^i}{E_o}(\mathbf{n}^i\cdot\sigma - \mathbf{n}^i\cdot\sigma\cdot\mathbf{n}^i\mathbf{n}^i) \qquad (2.18)$$

where $\beta = 16(1-\nu_o^2)/3\pi$ and $\gamma = \beta/(1-\nu_o/2)$. The compliances of a crack in normal and shear modes are *different*, $\beta \neq \gamma$ (unless $\nu_o = 0$), although not by much.

Substituting (2.18) into (1.5) (or substituting **B**-tensor 1.9 of a circular crack into 2.1,2) yields

$$\Delta f = \frac{8(1-\nu_o^2)}{3(1-\nu_o/2)E_o}\left\{(\sigma\cdot\sigma):\alpha - \right.$$

$$\left. -\frac{\nu_o}{2}\sigma : \frac{1}{V}\sum_i (l^3\mathbf{nnnn})^i : \sigma \right\} \qquad (2.19)$$

where

$$\alpha = \frac{1}{V}\sum_i (l^3\mathbf{nn})^i \qquad (2.5a)$$

is the crack density tensor (*3-D* equivalent of 2.5).

Unlike the 2-D case, Δf is *not* a function of σ and α *only*, but contains an *extra term* involving the fourth rank tensor $(1/V)\Sigma(l^3\mathbf{nnnn})^i$ that causes deviations from orthotropy. Emergence of this term (and, thus, insufficiency of α) is a consequence of *non-collinearity* of vectors $\langle \mathbf{b}^i \rangle$ and $\mathbf{n}^i\cdot\sigma$ in *3-D*. (Note that this term's contribution $\Delta\mathbf{M}''$ into the effective compliance has an unusual symmetry: since $(1/V)\Sigma(l^3\mathbf{nnnn})^i$ is symmetric with respect to *all* rearrangements of indices, $\Delta M''_{1122} = \Delta M''_{1212}$ and two similar relations, so that $\Delta\mathbf{M}''$ contributes equally into Poisson's ratios and the shear moduli).

However, *deviations from orthotropy* caused by the extra term are *small*, due to the following two factors.

1. The extra term has a relatively small multiplier $\nu_o/2$ (this is a consequence of a relatively small difference between normal and shear compliances of a circular crack). If its relative weight is evaluated by the ratio

$$\kappa = (\Delta M''_{ijkl}\,\Delta M''_{ijkl})^{1/2}/(\Delta M'_{ijkl}\,\Delta M'_{ijkl})^{1/2} \qquad (2.20)$$

of the Euclidean norms of the compliances $\Delta\mathbf{M}''$, $\Delta\mathbf{M}'$ corresponding to this term and to the α - term, then $\kappa < \nu_o/\sqrt{6}$ (Kachanov,1980a).

2. Only a *part* of this term causes deviations from orthotropy; a substantial part of $\Delta\mathbf{M}''$ has the structure of orthotropy coaxial to α and simply adds to the moduli $\Delta\mathbf{M}'$.

Examples below demonstrate smallness of errors caused by neglecting the extra term.

These errors can be further reduced if the normal and shear compliances of a crack are *enforced to be the same and equal to their average*, i.e. replacing β and γ in (2.18) by $(\beta+\gamma)/2$; then

$$\Delta f \approx \Delta f(\sigma,\alpha) = \frac{8(1-\nu_o^2)}{3E_o}\,\frac{1-\nu_o/4}{1-\nu_o/2}\,\sigma\cdot\sigma : \alpha \qquad (2.21)$$

and the fourth rank tensor is eliminated from the potential.

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Thus, characterization of a crack array by the crack density tensor α only constitutes a good approximation. The effective properties are then orthotropic.

A. *Isotropic (random) orientation distribution.* Both the crack density tensor α and the fourth rank tensor $(1/V)\Sigma(l^3\mathbf{nnnn})^i$ are isotropic. Since $tr\ \alpha = \rho$,

$$\alpha = (\rho/3)\ \mathbf{I} \tag{2.22}$$

($\mathbf{I}$ is a *3-D* unit tensor). Since tensor $(1/V)\Sigma(l^3\mathbf{nnnn})^i$, in addition to being isotropic, is symmetric with respect to all rearrangements of indices, its *ijkl*-components are: $C(\delta_{ij}\delta_{kl}+\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk})$. Since $(1/V)\Sigma(l^3\mathbf{n}\cdot\mathbf{nn}\cdot\mathbf{n})^i = \rho$ and $\delta_{ii}\delta_{kk}+2\delta_{ik}\delta_{ik}=15$, the constant $C = \rho/15$. Inserting this tensor and (2.22) into (2.19) yields

$$\Delta f = \frac{8(1-\nu_o^2)}{9(1-\nu_o/2)E_o}\rho[(1-\frac{\nu_o}{5})\sigma_{ij}\sigma_{ji} - \frac{\nu_o}{10}(\sigma_{kk})^2] \tag{2.23}$$

and the effective moduli readily follow:

$$E = E_o\,[1+\frac{16(1-\nu_o^2)(1-3\nu_o/10)}{9(1-\nu_o/2)}\rho]^{-1};$$
$$G = G_o\,[1+\frac{16(1-\nu_o)(1-\nu_o/5)}{9(1-\nu_o/2)}\rho]^{-1} \tag{2.24}$$

Note that *if only the α - term is retained* in (2.19), the effective moduli (2.24) change to

$$E = E_o\,[1+\frac{16(1-\nu_o^2)}{9(1-\nu_o/2)}\rho]^{-1};$$
$$G = G_o\,[1+\frac{16(1-\nu_o)}{9(1-\nu_o/2)}\rho]^{-1} \tag{2.25}$$

which are close to the rigorous result (2.24).

If the potential (2.21), obtained by replacing β and γ by $(\beta+\gamma)/2$, is used, then

$$E = E_o\,[1+\frac{16(1-\nu_o^2)(1-\nu_o/4)}{9(1-\nu_o/2)}\rho]^{-1};$$
$$G = G_o\,[1+\frac{16(1-\nu_o)(1-\nu_o/4)}{9(1-\nu_o/2)}\rho]^{-1} \tag{2.26}$$

which are even closer to the rigorous result (2.24).

B. *Parallel cracks* (normal to the x_1 axis). The crack density tensor

$$\alpha = \rho\ \mathbf{e}_1\mathbf{e}_1\ ; \tag{2.27}$$

the fourth rank tensor $(1/V)\Sigma(l^3\mathbf{nnnn})^i = \rho\mathbf{e}_1\mathbf{e}_1\mathbf{e}_1\mathbf{e}_1$. Thus,

$$f = f_o + \frac{8(1-\nu_o^2)}{3(1-\nu_o/2)E_o}\rho\,[\sigma_{1j}\sigma_{j1} - \frac{\nu_o}{2}\sigma_{11}^2] \tag{2.28}$$

so that

$$E_1 = E_o\,[1+\frac{16(1-\nu_o^2)}{3}\rho]^{-1};$$
$$G_{12} = G_o\,[1+\frac{8(1-\nu_o)}{3(1-\nu_o/2)}\rho]^{-1} \tag{2.29}$$

If only the α - term is retained in (2.19), then the moduli (2.29) change to

$$E_1 = E_o\,[1+\frac{16(1-\nu_o^2)}{3(1-\nu_o/2)}\rho]^{-1};$$
$$G_{12}\ \textit{remains unchanged} \tag{2.30}$$

or, if the potential (2.21) is used,

$$E_1 = E_o\,[1+\frac{16(1-\nu_o^2)(1-\nu_o/4)}{3(1-\nu_o/2)}\rho]^{-1};$$
$$G_{12} = G_o\,[1+\frac{8(1-\nu_o)(1-\nu_o/4)}{3(1-\nu_o/2)^2}\rho]^{-1} \tag{2.31}$$

Both (2.30) and (2.31) are close to the exact moduli (2.29).

C. To illustrate the orthotropic character of anisotropy in the cases of *more complex crack orientation statistics*, we consider an example (similar to (2.17) in *2-D*) of *two families of parallel cracks inclined at 30° to each other, one having the density twice as high as the other.* The crack density tensor is given by (2.17) and the fourth rank tensor $(1/V)\Sigma(l^3\mathbf{nnnn})^i = (2\rho/3)\mathbf{n}^1\mathbf{n}^1\mathbf{n}^1\mathbf{n}^1 + (\rho/3)\mathbf{n}^2\mathbf{n}^2\mathbf{n}^2\mathbf{n}^2$. The principal axes of α are the same as in the *2-D* case.

Unlike the *2-D* case, orthotropy, strictly speaking, does not hold, due to the fourth rank tensor's contribution. However, *the deviation from orthotropy is small.* Indeed, calculated in the principal axes of α, the effective compliances ΔM_{ijkl} (obtained by expressing α and the fourth rank tensor in the the principal axes of α and inserting them into 2.19) are as follows (normalized to ρ/E_o; $\nu_o = 0.25$ is assumed):

$$\begin{array}{llll} \Delta M_{1111} = 0.167 & \Delta M_{2222} = 2.370 & \Delta M_{3333} = 0 & \\ \Delta M_{1122} = -0.019 & \Delta M_{2233} = 0 & \Delta M_{3311} = 0 & (2.32) \\ \Delta M_{1212} = 1.400 & \Delta M_{2323} = 0 & \Delta M_{3131} = 0 & \end{array}$$

The rest of the moduli would have been exactly zeros in the case of rigorous orthotropy (the fourth rank tensor's contribution neglected). They are quite small, as compared to (2.32): the only non-zero ones are

$$\Delta M_{1112} = 0.004\ ,\quad \Delta M_{2212} = -\,0.004 \tag{2.33}$$

Their relative weight, evaluated by the ratio (2.20), is negligible: $\kappa = 1.4\times10^{-3}$.

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Note that, as seen from (2.32), ΔM_{1122} is of the order of only 1% of ΔM_{1212} and ΔM_{2222}, i.e. the impact of *cracks on Poisson's ratio* (in the principal axes of α) *is small*, as compared to the impact on the shear and Young's moduli. This illustrates the accuracy of representation of an arbitrary crack array by three mutually normal crack systems (and of relations 2.34 following from such representation).

Simplified character of orthotropy. Number of independent constants. As shown above, the effective elastic properties of *3-D* cracked solids are approximately orthotropic (orthotropy axes being coaxial with α).

Similarly to the *2-D* case, *the orthotropy is of a simplified type*:

(1) Directional variation of the compliances is described by *ellipsoids* (rather than surfaces of the fourth order, as in the case of general orthotropy).

(2) *The number of independent constants is drastically reduced, from nine* (general orthotropy) *to only four*, due to the following simplifications (similar to the ones in *2-D*):

(a) shear moduli can be expressed in terms of the Young's moduli and Poisson's ratios through equation of the type of (2.9a);

(b) under uniaxial loading in one of the principal directions of α, *the presence of cracks does not cause any additional Poisson's effect*:

$$\nu_{12}/E_1 = \nu_{23}/E_2 = \nu_{31}/E_3 \qquad = \nu_o/E_o \qquad (2.34)$$

This fact, together with the general symmetry relations $\nu_{ij}/E_i = \nu_{ji}/E_j$, implies that all six Poisson's ratios can be expressed in terms of the Young's moduli E_i and the matrix constant ν_o/E_o. It further implies (in conjunction with 2.9a) that the shear moduli, too, can be expressed in terms of E_i and ν_o/E_o:

$$1/G_{ij} = 1/E_i + 1/E_j + 2\nu_o/E_o \qquad (2.35)$$

Thus, *the entire set of elastic constants can be expressed in terms of three Young's moduli E_i (in the principal directions of α) and one matrix constant ν_o/E_o.*

These conclusions may have implications for *wave propagation in cracked media*, since they may allow one to interrelate the shear- and compressional wavespeeds in a medium with crack-induced anisotropy.

"Natural" coordinate system for effective elastic moduli. Plane problem. As discussed above, the effective elastic properties of solids with non-interacting cracks possess an orthotropy coaxial with α (exactly, in *2-D* and with good accuracy, in *3-D*).

The principal axes of α constitute, therefore, a "natural" coordinate system for calculation of the effective moduli. In this system, the matrix M_{ijkl} has its simplest (orthotropic) form, either exactly (in *2-D*) or approximately (in *3-D*).

Results of computer experiments on sample arrays (section 9) indicate that orthotropic symmetry still holds with satisfactory accuracy for *interacting* cracks, so that the mentioned "natural" coordinate frame remains the system of choice for computing M_{ijkl}.

This discussion has implications for the *plane problem of elasticity* for a cracked solid. In the case of an anisotropic solid, the plane problem can be formulated only in the plane of elastic symmetry. This implies that *plane problem for a cracked solid can be formulated only in one of the principal planes of the "natural" coordinate system.*

On the effect of dimensionality. Comparison of the stiffness reduction due to cracks in *2-D* and *3-D* (coefficients at ρ in formulas 2.13 and 2.24, or in formulas 2.16 and 2.29) shows that, at the same crack density, the reduction is *substantially weaker in 3-D.* This conclusion is somewhat relative, though, since each of the definitions (1.1) of ρ can always be made to incorporate an arbitrary multiplier. A somewhat more solid comparison can be based on adding multipliers π and $4\pi/3$ into the definitions of ρ in *2-D* and *3-D*, respectively, making ρ a relative area (volume) swept by cracks when they are rotated about their centers (diameters). Comparison based on thus modified ρ still shows a substantially weaker sensitivity to cracks in *3-D*: for randomly oriented cracks, fro example, the coefficient at ρ in the formula for E/E_0 in *3-D* constitutes only 0.42 0f the one in *2-D* (at ν_0=0.25).

We note, in this connection, that not only the impact of non-interacting cracks on the effective moduli is weaker in *3-D*, but *interactions between cracks* are also weaker in *3-D* (Kachanov and Laures,1989).

Non-circular cracks. We consider now *elliptical* crack shapes and analyze the impact of the shape factor on the overall effective properties.

As discussed in the Introduction, the crack compliance tensor for an elliptical crack has the form

$$\mathbf{B} = \frac{\eta+\zeta}{2}\left[\mathbf{I} + \left(\frac{2\xi}{\eta+\zeta} - 1\right)\mathbf{nn} + \frac{\eta-\zeta}{\eta+\zeta}(\mathbf{ss}-\mathbf{tt})\right] \qquad (2.36)$$

where $\boldsymbol{s}$ and $\boldsymbol{t}$ are unit vectors of ellipse's axes ($2a$ and $2b$) and ξ, η, ζ are given by elliptic integrals.

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An important factor for anisotropy of the overall properties is whether the vector <b> is collinear to $\mathbf{n}\cdot\boldsymbol{\sigma}$. From this point of view, the three terms of (2.36) play different roles.

The first term characterizes the part of <b> that is collinear to the traction vector $\mathbf{n}\cdot\boldsymbol{\sigma}$.

The second term characterizes deviation from this collinearity. Since $(\eta+\zeta)/2$ is the shear compliance averaged between the two principal directions of the ellipse, the coefficient at the second term is small if $(\eta+\zeta)/2$ is close to the normal compliance (as in the case of a circular crack).

The third term describes situation in the crack plane. It characterizes deviation from collinearity of the in-plane component of $\mathbf{n}\cdot\boldsymbol{\sigma}$ from the resulting in-plane component of <b> and vanishes if the crack is circular.

Substituting (2.36) into the general expression (1.5) yields the elastic potential as a sum of three terms corresponding to the three terms in (2.36):

$$\Delta f = \frac{\pi}{2}(\sigma\cdot\sigma): \frac{1}{V}\sum_i \left\{ ab\,\frac{\eta+\zeta}{2}\,\mathbf{nn}\right\}^i$$

$$+ \frac{\pi}{2}\,\sigma : \frac{1}{V}\sum_i \left\{ ab\,[\xi - \frac{\eta+\zeta}{2}]\mathbf{nnnn}\right\}^i : \sigma$$

$$+ \frac{\pi}{2}\,\sigma : \frac{1}{V}\sum_i \left\{ ab\,\frac{\eta-\zeta}{2}\,\mathbf{n(ss-tt)n}\right\}^i : \sigma \qquad (2.37)$$

The third term consists of a sum of the differences $\sigma_{ns}^2-\sigma_{nt}^2$ where σ_{ns}, σ_{nt} are the shear traction components along the *s*- and *t*-directions (of the major and minor ellipses axes). These differences enter the sum with multipliers that depend on the crack sizes and aspect ratios. If the aspect-ratio distribution is *uncorrelated* with the size and orientation distributions, then the third term vanishes.

The first two terms of (2.37) have the same general structure as the potential (2.19) of a solid with *circular* cracks. They are indeed very close to (2.19); this is seen by introducing equivalent circular cracks, as follows.

Guided by the crack density parameter (1.1a) - Budiansky and O'Connell's generalization of the parameter (1.1) for circular cracks - we replace the *i*-th crack contribution into (2.37) by the contribution of a *circular* crack having the same ratio S^2/P. The radius l of such an equivalent crack is then expressed in terms of the semiaxes a, b as $l^3 = \pi ab^2/(2E(k))$.

The elliptic crack and the equivalent circular crack produce exactly the same *normal* strain, i.e. have the same product (crack area)$\langle b_n\rangle$. The *shear* compliance of the equivalent circular crack *differs* from the averaged (between two principal directions of the ellipse) shear compliance of the elliptical crack. This difference depends on the aspect ratio *a/b* of the ellipse, but remains *small* even at large *a/b*: it is 0.7%, 2.5% and 3.2% for *a/b* =2, 6 and 58, respectively Therefore, the *vector* (crack area)$\langle b_\tau\rangle$ is approximately the same for these cracks, in terms of both the direction and the magnitude of this vector.

Thus, replacement of an elliptical crack by the equivalent circular crack is fully adequate. Potential (2.37) can be replaced, with good accuracy, by the potential (2.19) of a solid with equivalent *circular* cracks. Therefore, all the conclusions on orthotropy, descriptions in terms of crack density tensor, etc. hold for a solid with elliptical cracks.

This conclusion holds *provided* the aspect-ratio, size and orientation distributions of elliptical cracks are *uncorrelated* (so that the third term in 2.37 can be neglected); otherwise, the third term's contribution should be taken into account.

Historical remarks on crack density tensor, orthotropy and potentials in terms of invariants. Crack density tensor and construction of the elastic potential f in terms of simultaneous invariants of $\boldsymbol{\sigma}$ and $\boldsymbol{\alpha}$ were introduced by Vakulenko and Kachanov (1971). The crack density tensor was introduced there in a more general form that covers dislocations as well and, in the case of cracks, allows one to incorporate information on crack faces' displacements. Orthotropy of the effective properties was pointed out by Kachanov (1972). For the study of effective elastic properties of a solid with narrow cracks, (2.5-2.5a) is the appropriate definition; the corresponding discussion (with an overview of modifications and applications of the crack density tensor) was given by Kachanov (1980a). Such constructions of f in terms of invariants of $\boldsymbol{\sigma}$ and $\boldsymbol{\alpha}$ were used by Sayers and Kachanov (1991) as a tool for transforming the results for randomly oriented and parallel cracks produced by the self-consistent and differential schemes to the *arbitrary* orientation statistics (see section 8).

Tensors similar to $\boldsymbol{\alpha}$ and construction of the potential (or, equivalently, of the compliance tensor) in terms of invariants were subsequently used by several authors.

Oda et al (see, for example, 1984) used a "crack tensor" $\mathbf{F}$ coinciding with $\boldsymbol{\alpha}$ to within a multiplier 2π (the fourth rank fabric tensor was also used; however, it was additionally assumed that vectors $\mathbf{n}\cdot\boldsymbol{\sigma}$ and <b> are collinear in the *3-D* case - as discussed above, this assumption actually *eliminates* the fourth rank tensor's contribution into the effective moduli).

Litewka (see, for example, 1986) considered, primarily, periodic crack arrangements and used a symmetric second rank "damage tensor" $\mathbf{D}$ that differs from $\boldsymbol{\alpha}$ by carrying information on crack widths (as a consequence, the effective compliances may not be fully consistent with the rigorous results for narrow non-interacting cracks, see his formulas (5-6) at $\tau \to 0$).

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Krajcinovic (1985) used a similar construction of the potential in terms of invariants, with the difference that *vectors*, rather than tensors, were used for characterization of cracks (problems associated with vectorial characterization are discussed below). Talreja (1985) applied this construction (with vectors, 1985, changed to tensors in 1987) to the case of cracks in *anisotropic* matrix (see discussion of section 3).

Tensors of the α - type were applied by a number of authors to studies of various *fracture* processes.

The more general version of the crack density tensor that carries information on crack openings (Vakulenko and Kachanov, 1971, Kachanov, 1972) was applied by Allen et al (1987) and by Lee et al (1989) to the description of damage in composites, by Dragon and Mroz (1979) and by Dragon and Chihab (1985) and Chihab and Dragon (1990) in a study of damage evolution in rock-like materials and in ductile materials, by Talreja (1991) in a study of damage in ceramic composites, etc.

Murakami and Ohno (1981) used a "damage tensor" Ω which differs from the crack density tensor in being defined in terms of cross-sectional area loss (due to grain boundary cavitation) for description of anisotropic creep. General discussion of such area loss-related tensors was given by Betten (1982). Both Betten and Murakami and Ohno proposed constitutive laws in terms of simultaneous invariants of σ and the damage tensor (similarly to construction of elastic potential in terms of α and σ).

We note, as a general comment, that, whereas the area loss-related tensors and other modifications of the α - type tensors may be appropriate for characterization of various *fracture* processes, their use in the problem of *elastic* properties may lead to inconsistencies (use of the area loss-related tensors imply, for example, that the contribution of a given crack into the change of the overall property is proportional to its area, whereas it should be proportional to $(\text{area})^{3/2}$). For the *elastic* properties, tensor (2.5-2.5a) - naturally emerging in the elastic potential - should be used.

Applications of crack density tensor to other physical properties. Tensors of the α - type can be applied to characterization of other physical properties of cracked solids.

These tensors should then be *appropriately modified* - a multiplier at an individual crack's dyad **nn** should be changed from l^2 or l^3 (as for the elastic properties in *2-D* or *3-D*) to the one that properly reflects the *relative contribution of a given crack into the overall property.*

Analyses of this kind can be done, for example, for *fluid filtration* through a fissured medium and for the *electro- and heat conductivity* of a medium with crack-like isolators or conductors. Since these physical laws interrelate *vectorial* quantities, the tensors that characterize the physical property (and that have to be specified as functions of α) are of the *second* (rather than fourth) rank. The analysis is therefore simpler than in the case of elasticity.

To illustrate the logic of such analyses, we consider *fluid filtration* through a fissured material following the work of Kachanov, 1975, 1980a (see also a similar work of Oda,1985; for an earlier analysis of filtration anisotropy due to several planar families of fissures, that does not use tensorial representations and is lengthier, see Romm and Pozinenko,1963).

Darcy's law for the effective anisotropic permeability, as formulated by Ferrandon (1948), is

$$\mathbf{v} = -\mu^{-1}\,\mathbf{K}\cdot\nabla p \tag{2.38}$$

where $\mathbf{v}$ is the filtration rate, μ - viscosity coefficient and ∇p - externally applied pressure gradient. Symmetric second rank tensor $\mathbf{K}$ characterizes the anisotropy of filtration.

Assuming that the "background" permeability in the absence of fissures is isotropic ($\mathbf{K}=k\mathbf{I}$), we aim at specifying the change in permeability $\mathbf{K}-k\mathbf{I}$ due to fissures as a function of the α - type parameter. Tensor α should, however, be modified, to account for the fact that contribution of a given fissure into $\mathbf{v}$ is proportional to its area and to its opening cubed. Thus, α should be changed to

$$\alpha = (1/V)\sum_i (S\, b^3\, \mathbf{n}\,\mathbf{n})^i \tag{2.39}$$

Dependence of $\mathbf{K}-k\mathbf{I}$ on α is an *isotropic function* (if both ∇p and the array of fissures undergo any orthogonal transformation, say, rotation, then $\mathbf{v}$ undergoes the same transformation). The Cayley-Hamilton's theorem then implies that $\mathbf{K}-k\mathbf{I}$ is a polynomial quadratic in α, with coefficients-functions of its invariants. Since, for non-interacting flows, $\mathbf{v}$ is a sum of the individual fissures' contributions, the polynomial should be linearized in α (similarly to linearity of the elastic compliance in α in the case of non-interacting cracks). We further note that a family of parallel fissures does not affect permeability in the direction normal to them. These considerations allow one to fully determine $\mathbf{K}-k\mathbf{I}$ as a function of α:

$$\mathbf{K} - k\mathbf{I} = C\,[(\mathrm{tr}\,\alpha)\mathbf{I} - \alpha] \tag{2.40}$$

where C is a constant of a hydrodynamic nature.

Similar analysis can be applied to *electric (or heat) conductivity* of a medium with crack-like conducting or isolating surfaces, by specifying the corresponding tensor in Ohm's (or Fourier's) law as a function of the (appropriately modified) crack density tensor (Kachanov, 1976, 1980a).

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Note that dyads of the **nn**-type, formally similar to those comprising α, have been used in the (unrelated to cracks) problem of *contact stresses in a granular material* by Satake,1978 (see also review of Mehrabadi et al, 1982).

Non-coaxiality of different physical properties of cracked solids. Tensors characterizing anisotropy of elastic compliance, of fluid filtration, of electric conductivity, of heat conduction can all, in the approximation of non-interacting cracks, be specified as functions of symmetric second rank crack density tensors. One may be tempted, therefore, to use the measured anisotropy of one property (say, electric conductivity) to judge anisotropy of another property (say, elastic moduli or fluid permeability); such correlations are sometimes used in geophysics.

However, such correlations cannot generally be established, since tensors characterizing different properties are functions of *different modifications* of the crack density tensor. For example, the impact of fissures on fluid permeability is highly sensitive to the fissures' openings (as reflected in the definition 2.39 of the crack density tensor) whereas the linear elastic moduli are unaffected by them (as reflected in the definition 2.5-2.5a). Principal axes of these modifications of α and, therefore, *the principal axes of different physical properties have different orientations.*

Remark on characterization of cracks by vectors. Several authors (Talreja, 1985, Krajcinovic,1985, Costin, 1985) have suggested to characterize crack arrays by *vectors* (rather than second rank tensors); for a given crack, the vector is taken in the direction of a unit normal **n** and its length is determined by the crack size.

As is known, only *even rank* tensors may be used as such parameters (for the general discussion, see, for example, Onat and Leckie, 1988). We point out several inconsistencies resulting from the vectorial characterization.

1. It is not clear how to introduce the *overall* (average) vectorial parameter characterizing the entire crack array. In particular, it is not clear what such parameter would be in the simplest case of randomly oriented cracks (for a second rank tensor, no such difficulty arises - it would be proportional to a unit tensor **I**).

2. This difficulty is usually avoided by "discretization", i.e., by retaining the entire set of vectorial parameters $\mathbf{n}^i$ for each crack (or each family of parallel cracks) and using them as parameters in the elastic potential f, alongside with the stress tensor σ (or strain tensor ε). Then, f is constructed in terms of simultaneous invariants of **n**'s and σ (similarly to constructing of f in terms of invariants of σ and α).

However, one still runs into difficulties:

(a) the number of invariants that should be allowed into f is very large: among them are all the combinations $(\mathbf{n}^i\cdot\sigma\cdot\mathbf{n}^j)(\mathbf{n}^k\cdot\sigma\cdot\mathbf{n}^l)$ and $(tr\sigma)(\mathbf{n}^i\cdot\sigma\cdot\mathbf{n}^j)$, each entering f with a coefficient to be determined (for example, for three systems of parallel cracks there would be 42 coefficients).

(b) formation of simultaneous invariants requires that σ, as a second rank tensor, contracts *twice* with vectors $\mathbf{n}^i$. Thus, *there are no terms linear in* **n**'s. Therefore, the simplest case - *the approximation of non-interacting cracks -cannot be recovered.*

Concluding remark. The underlying reason for adequacy of the crack density tensor as a parameter of a crack field is *collinearity of vectors* $\boldsymbol{n}\cdot\sigma$ *and* $\langle\boldsymbol{b}\rangle$ (exact, in *2-D*, and approximate, in *3-D*). This implies orthotropy.

Therefore, *crack density tensor becomes inadequate to the extent this collinearity is violated.* Several physically important situations of this kind are discussed in the next three sections.

3. NON-INTERACTING CRACKS IN ANISOTROPIC MATRIX.

On compliance tensor of a crack in anisotropic solid. Modes I and II are coupled - normal or shear traction on a crack produces *both* normal and shear COD's.

This is reflected in non-diagonality of the **B**-tensor: $B_{tn}=B_{nt}\neq 0$. Also, **B=B(n)** - crack compliance depends on its orientation with respect to the anisotropy axes of the matrix.

For *arbitrarily oriented* cracks, **B**-tensors can be found only numerically (solving an algebraic equation, of the 4-th order in *2-D*, and of the 6-th order in *3-D*, is required). In this connection, the works of Kamel and Liaw (1989) and Tvardovsky (1990) on cracks in 2-D anisotropic solids may be useful.

One important case, when **B**'s and the effective moduli *can* be found in a closed form is considered below.

2-D orthotropic matrix with cracks parallel to the axes of orthotropy. In this case, the normal and shear modes are *uncoupled.* Using the solution of Ang and Williams (1961), we find the average normal and shear CODs for a crack normal to the x_2 axis:

$$\langle b_n\rangle=\frac{\pi l}{2}\left\{\frac{E_1^o}{G_{12}^o}-2\nu_{12}^o+2(E_1^o/E_2^o)^{1/2}\right\}^{1/2}(E_1^oE_2^o)^{-1/2}\sigma_{22}$$
$$= l\,B_{nn}^{(2)}\sigma_{22}$$
$$\langle b_\tau\rangle=\frac{\pi l}{2}\left[\left\{\frac{E_1^o}{G_{12}^o}-2\nu_{12}^o+2(E_1^o/E_2^o)^{1/2}\right\}^{1/2}/E_1^o\right]\sigma_{12}$$
$$= l\,B_{tt}^{(2)}\sigma_{12} \qquad (3.1)$$

where $B_{nn}^{(2)}$, $B_{tt}^{(2)}$ are components of the crack compliance tensor $\mathbf{B}^{(2)}$ (superscript 2 refers to the crack normal to the x_2 axis); $\mathbf{B}^{(2)}$ is diagonal ($B_{nt}^{(2)}=0$) and given by the formula

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$$\mathbf{B}^{(2)} = \frac{\pi l}{2}\left\{\frac{E_1^o}{G_{12}^o} - 2\nu_{12}^o + 2(E_1^o/E_2^o)^{1/2}\right\}^{1/2}(E_1^o E_2^o)^{-1/2} \times$$
$$\times [\mathbf{e}_2\mathbf{e}_2 + (E_2^o/E_1^o)^{1/2}\mathbf{e}_1\mathbf{e}_1] \qquad (3.2)$$

For a crack normal to the x_1 axis, $\mathbf{B}^{(1)}$ is obtained from $\mathbf{B}^{(2)}$ by interchanging indices 1,2 in the formulas above.

Introducing tensors $\mathbf{B}^{(1)}$, $\mathbf{B}^{(2)}$ into (2.2_1) we obtain the compliance ΔM due to two families of cracks normal to x_1 and x_2 (crack densities are ρ_1, ρ_2, respectively):

$$\Delta M = \rho_1 \mathbf{e}_1\mathbf{B}^{(1)}\mathbf{e}_1 + \rho_2 \mathbf{e}_2\mathbf{B}^{(2)}\mathbf{e}_2 \qquad (3.3)$$

(to be appropriately symmetrized) from which all the moduli follow; for example,

$$\frac{E_2^o}{E_2} = 1 + \pi\rho_2\left\{\frac{E_1^o}{G_{12}^o} - 2\nu_{12}^o + 2(E_1^o/E_2^o)^{1/2}\right\}^{1/2}\left\{\frac{E_2^o}{E_1^o}\right\}^{1/2}$$

$\frac{E_1^o}{E_1}$ *is obtained from* $\frac{E_2^o}{E_2}$ *by interchanging indices* 1,2

$$\frac{G_{12}^o}{G_{12}} = 1 + 2\pi G_{12}^o\left\{\rho_1 \frac{1}{E_2^o}\left[\frac{E_2^o}{G_{12}^o} - 2\nu_{21}^o + 2(E_2^o/E_1^o)^{1/2}\right]^{1/2}\right.$$
$$\left. + \rho_2 \frac{1}{E_1^o}\left[\frac{E_1^o}{G_{12}^o} - 2\nu_{12}^o + 2(E_1^o/E_2^o)^{1/2}\right]^{1/2}\right\} \qquad (3.4)$$

In the case of *one* system of parallel cracks, these formuli recover the ones of Gottesman et al, 1980 (which contain a misprint in the expression for shear modulus making it dimensionally incorrect).

Formulae (3.3) may serve as a convenient starting point for calculations of the *self-consistent and differential schemes*, by changing the matrix' material moduli to the effective ones in the expression for the **B**-tensor. Such calculations were done, in a slightly different way, by Gottesman et al, 1980 (self-consistent scheme) and Hashin, 1988 (differential scheme).

The influence of various moduli of the orthotropic matrix on the compliance change due to cracks (aligned with the orthotropy axes) was analyzed in detail (with crack interactions taken into account) by Mauge (1990) and Mauge and Kachanov (1990, 1992).

The impact of cracks on stiffness reduction in the direction normal to them is stronger (or weaker), as compared to the case of *isotropic* matrix, if the matrix material is stiffer (or softer) in this direction. This effect is "asymmetric": the deviation from the case of isotropy is larger in the case of cracks normal to the *stiffer* direction.

Consider a numerical example (relevant for glass fiber reinforced plastics): $E_1^o/E_2^o = 4.1$; $E_1^o/2G_{12}^o = 5$; $\nu_{21}^o = 0.068$; $\nu_{12}^o = 0.277$. For parallel cracks normal to the "softer" direction, we obtain $E_2/E_2^o = (1+5.77\rho_2)^{-1}$ - a small change, as compared to $E/E^o = (1+2\pi\rho)^{-1}$ for the *isotropic* matrix with parallel cracks. For cracks normal to the "stiffer" direction, the difference with the case of isotropy is much more pronounced: $E_1/E_1^o = (1+11.5\rho_1)^{-1}$. Thus, the *anisotropic matrix is much more "sensitive" to cracks normal to the stiffer direction.*

Similar conclusion was reached by Mauge (1990) and Mauge and Kachanov (1990,1992) for *interacting* cracks in an orthotropic matrix: if cracks are normal to the stiffer (softer) direction, then interactions are enhanced (weakened), as compared to the case of an isotropic matrix.

Note that, although the overall anisotropy in the considered case is *orthotropic* (as follows from the structure of 3.5 and diagonality of **B**'s), ΔM *cannot* be expressed in terms of α - it depends on tensors $\mathbf{B}^{(1)}$, $\mathbf{B}^{(2)}$ containing information on matrix moduli.

On formulations in terms of invariants. For cracks *arbitrarily oriented* in an anisotropic matrix, **B**-tensors can be found only numerically, and progress beyond the general representation (2.2) becomes difficult.

As an alternative, one can take a "phenomenological" approach and assume that, similarly to the case of the isotropic matrix, the potential f is a function of σ and α but they enter f through terms that are invariant *only with respect to those orthogonal transformations that belong to the group of elastic symmetry of the matrix.* This approach is by no means rigorous, since there is no justification for excluding from the analysis the following factors (the only reason for disregarding them is that otherwise the analysis becomes intractable):

(1) the fourth rank fabric tensor (particularly because the difference in normal and shear compliances of a crack - the main reason for emergence of such a tensor in the potential - may be significant in the case of anisotropic matrix), and

(2) **B**-tensors, containing information on the matrix moduli.

Then the group of the effective elastic symmetry is an intersection of the groups of symmetry of the matrix and of α (the latter has symmetry of an ellipsoid); for example, an orthotropic matrix with one arbitrarily oriented family of parallel cracks will have no elements of elastic symmetry (and will require a full set of 21 elastic constants).

Expressions for f will then contain a large number of invariants, each entering with a coefficient to be determined (with the hope that information on the matrix moduli that should, *in a proper way*, enter f through **B**-tensors, can instead be incorporated through these coefficients - an additional assumption of unclear validity). Such an approach was briefly outlined by Kachanov (1980a); expressions for several cases of the matrix' symmetry were written down by Talreja (1987) (in the earlier paper, Talreja, 1985, *vectors* were used for characterization of crack arrays; inconsistencies associated with such characterization are discussed above).

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This approach leads to emergence of a large number of coefficients at invariants that have to be found experimentally. Since writing f in the form $f(\sigma, \alpha)$ cannot be rigorously justified in the case of anisotropic matrix, these coefficients are not material constants and may simply play the role of adjustable parameters (although Talreja's data seem to indicate their constancy in certain experiments).

4. NON-INTERACTING CRACKS CONSTRAINED AGAINST OPENING (BUT ALLOWED TO SLIDE)

This case is relevant for the compressive conditions. The difference in the effective properties between this case and the case of freely opening cracks accounts for the differences in compliance under compressive and tensile conditions.

Since the deviation from collinearity of the vectors $\mathbf{n}\cdot\sigma$ and $\langle\mathbf{b}\rangle$ is substantial ($\langle\mathbf{b}\rangle$'s have only tangential components), the crack density tensor α becomes inadequate and *orthotropy does not hold.* The situation may be further complicated by friction. We assume here that no secondary ("wing") cracks form.

The first quantitative study of a solid with frictionally sliding cracks under compression was done, probably, by Walsh (1965a,b) who found the effective Young's and bulk moduli. Stress-induced anisotropy, path-dependence of deformation and formation of a vertex on a loading surface were analyzed in detail by Kachanov (1980b, 1982). Self-consistent approximation was applied to the same problem by Horii and Nemat-Nasser (1983); however, as discussed in section 9, the self-consistent scheme may actually produce errors larger than the approximation of non-interacting cracks.

Sliding without friction ("lubricated cracks"). In this case the behaviour is *linear elastic* (in contrast with the case of cracks behaving differently in tension and compression). The potential can be found by introducing $\langle\mathbf{b}\rangle$'s (tangential to cracks) into (1.5).

Thus, in both *2-D* and *3-D* cases (circular cracks), $\langle\mathbf{b}\rangle$ is proportional to the shear traction $\mathbf{n}\cdot\sigma - \mathbf{n}\cdot\sigma\cdot\mathbf{nn}$ with the proportionality coefficient $\pi l/E_o'$ in *2-D* and $\gamma l/E_o$ in *3-D* (see 2.3 and 2.18). The potential takes the form

$$\Delta f = (\pi/E_o')\left\{(\sigma\cdot\sigma) : \alpha - \right.$$

$$\left. - \sigma : (1/A)\Sigma(l^2\mathbf{nnnn})^i : \sigma\right\} \quad \text{in } 2\text{-}D \quad (4.1)$$

and

$$\Delta f = (\gamma\pi/E_o)\left\{(\sigma\cdot\sigma) : \alpha - \right.$$

$$\left. - \sigma : (1/V)\Sigma(l^3\mathbf{nnnn})^i : \sigma\right\} \quad \text{in } 3\text{-}D \quad (4.2)$$

It has the form similar to (2.19) - crack density tensor α becomes insufficient and the fourth rank tensor $(1/A)\Sigma(l^2\mathbf{nnnn})^i$ emerges as a term in the potential. However, in contrast with (2.19), a *small multiplier* $\nu_o/2$ *at this term is absent*; therefore, *contribution of this term into the overall compliance can no longer be neglected.* (Unlike the case of freely opening cracks, the fourth rank tensor appears in both *2-D* and *3-D*). Therefore, *orthotropy is significantly violated.*

The case of *elliptical* cracks (with aspect ratio distribution uncorrelated with the orientational distribution), can be reduced to the case of equivalent circular cracks, the same way it was done for freely opening cracks.

Two main effects of the constraint are:

(1) it drastically reduces strains produced by cracks. In the *3-D* case of *randomly oriented* cracks, imposition of the constraint reduces the additional uniaxial compliance due to the presence of cracks by the factor of $0.4/(1-3\nu_o/10)$.

(2) In the case of *non-random* crack orientations, the reduction of strain is even more pronounced. Also, the constraint changes the direction in which the compliance is maximal, i.e. it *rotates the anisotropy axes*. This is a consequence of the tensor $(1/A)\Sigma(l^3\mathbf{nnnn})^i$ becoming important. The *2-D* case of parallel cracks provides an example (fig.2). The compliance is maximal for *unconstrained* cracks in the direction *normal* to cracks, whereas for *constrained* cracks the strain response in this case is zero, since shear displacements of crack faces - the only source of inelastic strains - are absent. Note that the maximal (over directions) compliance is reduced almost four times when the constraint is imposed.

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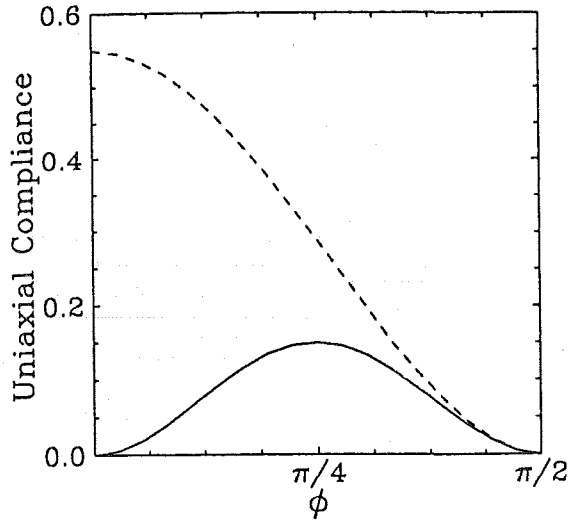


FIG.2. Uniaxial compliance (additional, due to cracks) of a *2-D* solid with parallel cracks, as a function of direction of loading (forming angle ϕ with a normal to cracks (normalized to $\pi p\sigma/E_0$). Solid and dashed lines indicate constrained and unconstrained cracks. Imposition of the constraint shifts the direction of maximal compliance by 45° and sharply reduces strains. This illustrates the difference between responses to compression and tension.

We note that the results above can be obtained by using the crack compliance tensor, changed from (1.8) to

$$\mathbf{B} = (\pi l/E_O') (\mathbf{I} - \mathbf{nn}) \tag{4.3}$$

in the 2-*D* case, and from (1.9) to

$$\mathbf{B} = \gamma(\mathbf{I} - \mathbf{nn}) \tag{4.4}$$

in the *3-D* case of circular cracks.

The results obtained above can be generalized to *elliptical* cracks by changing the **B**-tensor given by (1.10) to

$$\mathbf{B} = \eta \mathbf{ss} + \zeta \mathbf{tt} \tag{4.5}$$

Sliding with friction. If friction is present, vectors $\langle \mathbf{b}^i \rangle$ are not unique functions of stress, so that potential does not exist (the tensor of incremental compliance loses symmetry: $\Delta M_{ijkl} \neq \Delta M_{klij}$). Deformation is *loading path-dependent*, and *stress-induced anisotropy* emerges; *vertex* develops on the loading surface; unloading starts with a *"deadband"* (stress interval in which no backslidings occur).

These issues were investigated by Kachanov (1980b,1982); below, we briefly review the basic results.

"Driving force" that produces shear COD is the shear traction reduced by the frictional resistance. The simplest model for the latter is the Coulomb's law. Thus, the driving force on a crack is $\tau - \mu\sigma_n$ (τ and σ_n are the shear and normal tractions induced on a crack and μ is the friction coefficient); it depends on the crack orientation with respect to loading.

Loading path dependence is obvious from the observation that application of a shear traction τ (while holding $\sigma_n=0$) followed by application of compressive σ_n produces a non-zero shear COD, whereas reversing the sequence produces no displacement, if σ_n is sufficiently high. Because of the loading path dependence, stress-strain relations have to be formulated in the *incremental* form (similarly to the equations of metal plasticity).

However, in contrast with plasticity, the range of loading paths for which deformation is path-*independent* ("fully active" loadings - all cracks that started to slide, continue to do so, as loading proceeds) is *quite wide*. For example, in the case of axisymmetric compression

$$\sigma = -\, q\, (\mathbf{e}_1\mathbf{e}_1 + \mathbf{e}_2\mathbf{e}_2) - p\, \mathbf{e}_3\mathbf{e}_3$$

where $p>0$ and $q>0$ are the axial and lateral pressures, with a superimposed stress increment

$$d\sigma = -\, dq\, (\mathbf{e}_1\mathbf{e}_1 + \mathbf{e}_2\mathbf{e}_2) - dp\, \mathbf{e}_3\mathbf{e}_3$$

the range of dp,dq that provides fully active loading, is shown in fig. 3. This range is quite wide: at $\mu=0.6$, for example, the angle ONA differs from π by 18° only.

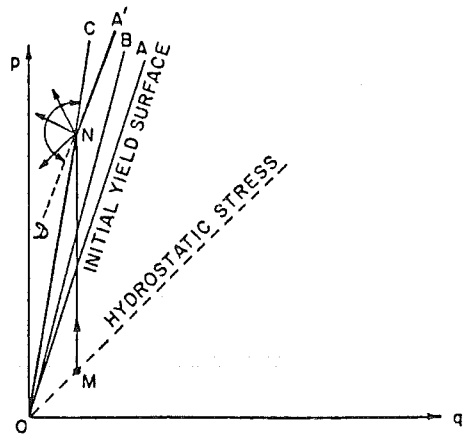


FIG.3. Range of path-independence (cone CD) and formation of a vertex on the loading surface in axisymmetrical compression; NA' is parallel to the initial loading surface OA most favourably oriented crack.MN shows the loading path (axial compression at constant lateral stress).

Note formation of a *vertex* on the loading surface (in qualitative terms, development of a vertex in a solid with frictionally sliding cracks was discussed by Rudnicky and Rice, 1975, where a phenomenological model was suggested). Vertex becomes more pronounced as loading proceeds. Strength of this effect depends on friction coefficient, increasing with μ and disappearing at $\mu=0$.

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Stress-induced anisotropy develops, as loading proceeds, since the strain increment produced by an incremental loading $d\sigma$ superimposed on a current state σ depends on orientation of $d\sigma$ with respect to σ: it is maximal if $d\sigma$ is in the direction of σ (maximal increase of driving forces $\tau-\mu\sigma_n$ on the cracks activated at σ). It is smaller for directions moderately deviating from σ (but still falling in the range of fully active loadings) and is even smaller (or zero) for $d\sigma$ forming large angles with σ, since they may "lock" the previously activated cracks.

Referring to Kachanov (1980b,1982) for the general expression for ΔM (anisotropy developing as a function of loading), we show in Table 1 a gradual development of stress-induced anisotropy in the *axisymmetric* case, as the ratio p/q increases. Incremental compliances are calculated in the assumption of $d\sigma$ belonging to the fully active range. Note a gradually increasing difference between ΔM_{1111} and ΔM_{3333} and that $\Delta M_{1133} \neq \Delta M_{3311}$ (indicating that, as expected for path-dependent deformation, potential does not exist).

TABLE I. Development of stress-induced anisotropy in the axisymmetrical loading, as the ratio p/q of the axial and lateral stresses increases. It is assumed that $\mu = 0.6$, so that frictional slidings start at p/q = 3.12. Compliances are normalized to $8\rho(1-\nu^2)/(3(2-\nu)E_0)$.

p/q	3.5	5.0	10
C_{1111}	0.190	0.340	0.458
C_{3333}	0.150	0.220	0.238
C_{3311}	- 0.241	- 0.382	- 0.439
C_{1133}	- 0.075	- 0.110	- 0.119

Note that, although the inelastic strains produced by frictional sliding are, typically, relatively small (as compared to the strains of the matrix), they may produce a distinguishable increase of the overall Poisson's ratio.

5. NON-INTERACTING CRACKS FILLED WITH COMPRESSIBLE FLUID

Collinearity of vectors $\mathbf{n}\cdot\sigma$ and $\langle \mathbf{b} \rangle$ is significantly violated, since the presence of fluid reduces the normal compliance of a crack-like cavity, whereas the shear compliance is assumed to be unaffected. Therefore, *orthotropy of the effective properties is noticeably violated.* Note that similar conclusions were reached, on somewhat different grounds, by Sayers(1988) who noted that the elastic anisotropy (as inferred from wave velocities) is reasonably well described by ellipsoids in *dry* rocks, but substantially deviates from such shapes in *fluid-saturated* rocks.

We consider here crack-like cavities filled with compressible fluid, in the approximation when stress interactions between cavities are neglected.

Our analysis is similar to the one of Budiansky and O'Connell (1976) in the way it considers coupling between stresses and fluid pressures (our derivation seems somewhat simpler). Their analysis was restricted to the case of randomly oriented cracks and was done in the framework of the self-consistent approximation; this approximation appears to substantially underestimate stiffness and may actually yield errors larger than the approximation of non-interacting cracks (see sections 8,9). Our results apply to the general orientational distribution; they also describe dependence of fluid pressures on cavities' orientation (pressure "polarization").

We assume the "undrained" approximation, i.e. that the total mass of fluid in each cavity remains constant (corresponds to a "short time" response in the case of fluid diffusion across crack faces).

For simplicity of calculations, we also assume that fluid-filled cavities have shapes of relatively thin "platelets". We first consider the case of cavities circular in plane. The representative cavity has radius l and, in the absence of applied stresses, has average opening $\langle b_0 \rangle$, average aspect ratio $\zeta_0 = \langle b_0 \rangle / l$ and volume $\pi l^3 \zeta_0$.

When stresses are applied, the cavity changes its volume, producing the change of fluid pressure Δq; the latter counteracts the applied traction, reducing the normal COD. Thus, fluid reduces the impact of cracks on the effective compliance, as compared to the case of *dry* cracks. This effect is somewhat similar to the impact of constraint against opening, but is milder ("soft" constraint).

Since the cavity is thin, its volume change is mainly due to changes of $\langle b_0 \rangle$ (l remains approximately constant). Thus, the relative change of cavity's volume $\Delta V/V_0 = \langle \Delta b_n \rangle / \langle b_0 \rangle$. Since the fluid mass within a cavity is constant ("undrained" conditions), the change of fluid density η is related to the volume change: $\eta_0 V_0 = \eta V$, or

$$\Delta\eta/\eta_0 = -\Delta V/V_0 = -\langle \Delta b_n \rangle / \langle b_0 \rangle \qquad (5.1)$$

The corresponding change of pressure is given by the constitutive equation for a compressible fluid, which, in the linearized formulation (small density changes), has the form

$$\Delta\eta/\eta_0 = -\kappa\,\Delta q \qquad (5.2)$$

where κ is the compressibility of a fluid (generally, a function of q and temperature; treated as constant here); in conjuction with (5.1), this equation relates Δq to the change of opening $\langle \Delta b_n \rangle$.

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On the other hand, $<\Delta b_n>$ is related to the overall normal traction on a cavity ($\mathbf{n}\cdot\sigma\cdot\mathbf{n} + \Delta q$) by a "cavity compliance" relation:

$$<\Delta b_n> = \beta l/E_o\ (\mathbf{n}\cdot\sigma\cdot\mathbf{n} + \Delta q) \tag{5.3}$$

where $\beta = 16(1-\nu_o^2)/3\pi$.

These equations yield normal COD's and changes of fluid pressures induced on cracks of different orientations by the applied stress σ:

$$<\Delta b_n> = \frac{l\beta}{E_o}\frac{\beta^{-1}\kappa E_o\zeta_o}{1+\beta^{-1}\kappa E_o\zeta_o}\ \mathbf{n}\cdot\sigma\cdot\mathbf{n} \tag{5.4}$$

$$\Delta q = -\frac{1}{1+\beta^{-1}\kappa E_o\zeta_o}\ \mathbf{n}\cdot\sigma\cdot\mathbf{n} \tag{5.5}$$

Thus, *sensitivity of fluid pressure in a given cavity to the applied stress is determined by a dimensionless constant*

$$\delta = \beta^{-1}\ \kappa E_o\zeta_o \tag{5.6}$$

This parameter - the key constant of the problem - is similar to the one used by Budiansky and O'Connell. It characterizes *coupling between stresses and fluid pressures* and determines the impact of fluid on the effective compliance.

Physically, the equation (5.3) means that the applied compressive stress is counteracted by the "cavity's stiffness" and by the fluid pressure. The parameter δ determines relative shares of these two factors.

In the limit of highly compressible fluid ($\kappa\to\infty$, "air"), change of the fluid pressures in cavities caused by applied stresses is zero and *the effective compliance is the same as in the case of "dry" cracks.* In the opposite limit $\kappa\to 0$ (incompressible fluid), $\Delta q = -\mathbf{n}\cdot\sigma\cdot\mathbf{n}$ and *compliance is the same as in the case of cracks constrained against opening*. The same is true if κ is finite but cavities are infinitesimally thin, $\zeta_o\to 0$, indicating that coupling between stresses and fluid pressures is stronger for thinner cracks.

In general, the compliance is *intermediate between the one of the "dry" medium and the one with cracks constrained against opening.*

As an example that may be relevant for fluid-saturated rocks, we consiser the case of $\kappa = 0.5\times10^{-3}\text{MPa}^{-1}$ (water) and $E_o = 6\times10^4$ MPa (granite). Then, for the aspect ratios $\zeta_o < 0.006$ the impact of fluid on the effective compliance is so strong that the compliance differs by less than 10% from the one of a medium with cracks constrained against opening. The impact of fluid becomes small (compliance differs from the one of a *dry* medium by less than 10%) only for very wide cavities, at $\zeta_o > 0.5$ - 0.6 (note that the calculation of cavity's volume change used above becomes then inaccurate). Thus, in the entire practically interesting range of $\zeta_o < 0.5$, the effect of stress-fluid pressure coupling is strong and significantly affects the effective compliance.

The elastic potential is obtained by introducing the normal COD's from (5.4) (and leaving the shear COD's unchanged) into the general expression (1.5). Thus,

$$\Delta f = \frac{8(1-\nu_o^2)}{3(1-\nu_o/2)E_o}\Big\{\sigma\cdot\sigma : \alpha - \Big[1-(1-\frac{\nu_o}{2})\times \times\frac{\delta}{1+\delta}\Big]\ \sigma : (1/V)\Sigma(l^3\mathbf{nnnn})^i : \sigma\Big\} \tag{5.7}$$

At $\kappa E_o\zeta_o\to\infty$, the expression (2.19) for dry cracks is recovered; at $\kappa E_o\zeta_o\to 0$, the expression (4.2) for cracks constrained against opening is recovered. In general, the coefficient at the term containing the fourth rank tensor is not small as compared to the one at the α -term, so that *deviations from orthotropy can be significant.*

Effective compliances follow from (5.7). For example, in the case of *randomly oriented* cracks,

$$(E/E_o)^{-1} = 1 + \rho\frac{16(1-\nu_o^2)}{9(1-\nu_o/2)}\Big\{1 - -\frac{3}{5}\Big[1-(1-\frac{\nu_o}{2})\frac{\delta}{1+\delta}\Big]\Big\}$$

$$(G/G_o)^{-1} = 1 + \rho\frac{16(1-\nu_o^2)}{9(1-\nu_o/2)}\Big\{1 - -\frac{2}{5}\Big[1-(1-\frac{\nu_o}{2})\frac{\delta}{1+\delta}\Big]\Big\} \tag{5.8}$$

Fig. 4 shows the Young's modulus as a function of crack density at various values of $\kappa E_o\zeta_o$.

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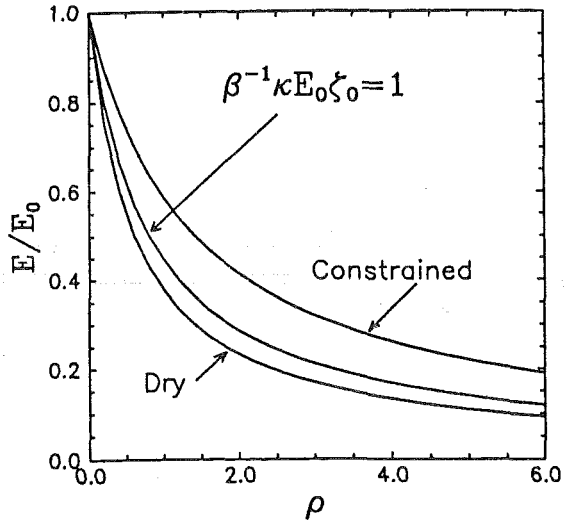


FIG.4. Young's modulus of a solid with fluid-filled cracks as a function of crack density, at various values of $\kappa E_0\zeta_0$. Note a gradual transition from the "dry" cracks curve to the curve for cracks constrained against opening.

Note that the results above can be obtained by using the crack compliance tensor of a circular crack, changed from (1.9) for a dry crack to

$$\mathbf{B} = \gamma\left(\mathbf{I} - \frac{\nu_0}{2}\frac{\delta}{1+\delta}\mathbf{nn}\right) \tag{5.9}$$

and that the results obtained above can be generalized to *elliptical* cracks by changing the **B**-tensor given by (1.11) to

$$\mathbf{B}=\frac{\eta+\zeta}{2}\left[\mathbf{I}+\left(\frac{2\xi}{\eta+\zeta}\frac{\delta}{1+\delta}-1\right)\mathbf{nn}+\frac{\eta-\zeta}{\eta+\zeta}(\mathbf{ss}-\mathbf{tt})\right] \tag{5.10}$$

Polarization of fluid pressure. Aside from the impact of fluid on the effective compliance, the result (5.5) describes *"pressure polarization"*: fluid pressure in a given cavity depends on the cavity's orientation with respect to the applied loading. In a medium with cavities of different orientations, fluid pressures will be different in different cavities.

It has been suggested (Rice, 1978) that a second rank "pressure polarization tensor" **Q** can possibly be introduced, such that a fluid pressure change in the cavity of orientation **n** is expressed as $\Delta q = \mathbf{n}\cdot\mathbf{Q}\cdot\mathbf{n}$. As seen from (5.6), such a tensor can be introduced *provided* all the cavities have the same aspect ratio ζ_0 - condition that may be too restrictive; if it *is* satisfied, then $\mathbf{Q} = -(1+\delta)^{-1}\sigma$. Note that $\mathbf{Q} = -\sigma$ in the limit of $\kappa E_0\zeta_0 \to 0$.

On drained and undrained conditions. The analysis above is relevant for the *undrained* conditions. If fluid diffuses across the cavities' faces, fluid pressures in different cavities gradually equalize and pressure polarization disappears. In the "long time" limit (*fully drained* response) the effective compliance becomes the same as the one of the "dry" medium. Thus, *in the process of fluid diffusion, the effective compliance changes from the one found above to the one of a dry medium* (given by formulae of section 2).

The case of *interacting* fluid-filled cavities is analyzed (on the basis of the technique outlined in section 9) by Lehner and Kachanov (1992).

6. SUMMARY ON NON-INTERACTING CRACKS.

For non-interacting cracks in an *isotropic* matrix, the results are exact for any orientational distribution of cracks, both in *2-D* and *3-D*, and are free of any controversy. They depend on the density and orientations of cracks but are independent of mutual positions of cracks.

As discussed in sections 7,9, this approximation may have a *wider than expected applicability*, due to cancellation of the competing effects of stress shielding and amplification in arrays of interacting cracks.

In the 2-D case, the second rank *crack density tensor* α provides a fully adequate characterization of a crack array. Linear invariant $tr\ \alpha = \rho$; thus, α is a tensorial generalization of ρ accounting for crack orientations. *In the 3-D case*, a fourth rank tensor should be used, alond with α. However, its impact on the effective moduli is small, so that characterization of cracks by α only is a good approximation

Although the compliances are linear in crack density, *the assumptions of non-interacting cracks and of small crack density are not equivalent* (see the beginning of section 2).

Since α is symmetric and enters the potential through a simultaneous invariant with σ, the effective moduli always possess *orthotropy* coaxial with the principal axes of α (this result is exact in *2-D* and approximate in *3-D*). Moreover, the orthotropy is of a simplified type - it is characterized by four (rather than nine) independent constants.

This gives rise to the concept of *"natural" coordinate system* for calculation of the effective moduli, in which the matrix of moduli is orthotropic in *2-D* and is close to such form in *3-D*. (For *interacting* cracks, as discussed in section 9, orthotropy still holds with good accuracy, so that the "natural" coordinate frame remains the system of choice).

Adequacy of α as a sole parameter of the crack field and orthotropy of the effective properties are based on *collinearity* of vectors $\langle\mathbf{b}\rangle$ and $\mathbf{n}\cdot\sigma$, i.e., on *equal (or close) compliances* of cracks in normal and shear modes.

If the collinearity is lost (cracks in anisotropic matrix, cracks constrained against opening and fluid-filled cracks), α becomes inadequate and the anisotropy is more complex.

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7. IMPACT OF INTERACTIONS ON THE EFFECTIVE MODULI: GENERAL REMARKS.

For *interacting cracks*, determination of effective modili becomes much more complex. In principle, its solution requires solving the interaction problem for each realization of the crack statistics and subsequent ensemble averaging, that comprises averaging over (a)orientations, (b)positions and (c)sizes of cracks. Since (a)-(c) are interdependent, this presents a very difficult (particularly in *3-D*) problem related to geometrical probabilities. A number of approximate schemes avoiding these difficulties were suggested in literature. Some of them are discussed in the next sections.

We first make several comments of a general nature.

The key question is: do the interactions *increase or reduce* the overall stiffness (as compared to the one for non-interacting cracks) ? The answer depends on whether stress *shielding* or stress *amplification* (resulting in stiffening or softening effects, correspondingly) *dominate* in a crack array.

We start with a general observation that, if tractions are prescribed at the boundary, then *introduction of cracks does not change the average stress* in the solid (see Introduction). Therefore, one may expect that the shielding and amplifying effects, in the average, *cancel* each other in an array of interacting cracks, so that the overall stiffness may be well described by the approximation of non-interacting cracks.

This argument is based on the assumption that there is *no "bias" in crack statistics towards either amplifying or shielding configurations*. Conditions leading to the "bias" may, to some extent, be understood by examining the structure of the stress field associated with *one* isolated crack (although such considerations are by no means rigorous since the field is distorted by the presence of other cracks).

An important observation is that, since the singularity zones near the crack tips provide a large contribution into the average stress, the zone where a given stress component has the opposite sign should be relatively *large* (to compensate for the singularity contribution).

As an example (relevant for parallel arrays under normal loading) fig.5 shows the field of the stress component σ_{yy} generated by a *2-D* crack subjected to a uniform tensile stress $\sigma_{yy}=1$. The field consists of small and intense amplification zones (near the tips) and more "diffused" shielding zones.

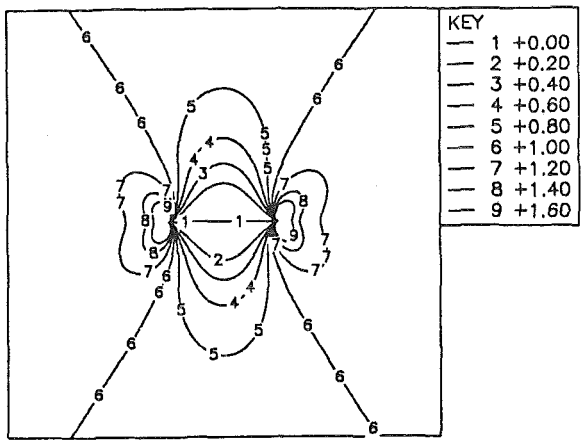


FIG.5. Zones of shielding and amplification of the stress component σ_{yy} generated by a crack subjected to a uniform tension (lines of equal level are indicated). The shielding zone is less intense but larger than the amplification zone.

This figure explains the fact that the (stress amplifying) *collinear* interactions become noticeable only at close spacings whereas the *stacked* interactions have a longer range. It also provides an insight into interactions in a rectangular doubly periodic array of parallel cracks, where the amplification effect dominates (stiffness is lower than in the approximation of non-interacting cracks) only if the ligaments between collinear cracks are substantially smaller than spacings between stacked cracks; otherwise, shielding effect is dominant (interactions produce stiffening).

For a *random array of parallel cracks* under normal loading p, the net result of interactions is determined by competition between the amplification effect generated by compact, but intensive at small spacings, zones of $\sigma_{yy} > p$ near crack tips and "diffused" shielding zones where $\sigma_{yy} < p$. A representative crack may thus experience a background shielding with superimposed local amplification zones. (Coupling of the modes will also take place: shear modes on cracks appearing due to stochastic asymmetries in a crack array will produce normal tractions along the other cracks' sites; it appears, however, that this effect - generally, weak - can be neglected, since, for randomly located cracks, shear tractions will have random signs and their impacts will cancel each other).

The outcome of this competition in an array of randomly located cracks may be expected to be a *cancellation* of the amplification and shielding effects, since the average stress is unaffected by the presence of cracks, $\langle\sigma_{yy}\rangle = p$.

Computer experiments on sample arrays (section 9) show that the effective stiffness is indeed close to the one predicted by the approximation of non-interacting cracks. At the same time, the stress intensity factors were substantially affected by interactions, indicating that interactions were strong.

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Thus, predictions of the approximation of non-interacting cracks remain accurate at high crack densities *not* because the interactions are weak, but because *the competing interaction effects cancel each other*. In view of conservation of the average stress when cracks are introduced (tractions prescribed on the boundary), this result is not surprising.

We emphasize that these arguments are based on the assumption that there is no "bias" in crack statistics towards either amplifying or shielding configurations. If crack locations *are* biased towards stacked (columnar) arrangements, then the shielding (stiffening) effect of interactions will dominate; if, on the other hand, the *amplifying* (collinear) interactions are dominant (widely separated rows of closely spaced collinear cracks), then the overall stiffness is *reduced* by interactions.

8. APPROXIMATE SCHEMES.

Some of the approximate schemes for the effective moduli are reviewed here. Most of them reduce the analysis to *one* isolated crack, placed into some sort of "effective" environment. They attempt to improve predictions of the approximation of non-interacting cracks; whether this is actually achieved, is discussed in section 9.

Self-consistent scheme (SCS). A representative inhomogeneity (inclusion, cavity, crack) is placed into a matrix with the *effective* elastic moduli, so that the impact of interactions on a given inhomogeneity is simulated by a *reduced stiffness of the surrounding material*. For a material with *cracks*, this scheme was developed by Budiansky and O'Connell (1976) for random crack orientations and applied by Hoenig (1979) to crack orientations that are either parallel or have cylindrical symmetry and by Horii and Nemat-Nasser to frictionally sliding cracks. It was further extended to 2-D orthotropic matrices with cracks parallel to the orthotropy axes by Gottesman et al (1980) and reformulated in a somewhat different form, with particular attention paid to crack shapes, by Laws and Brockenbrough (1987).

For illustration, we review only the case of random orientations, when the effective properties are isotropic.

We recall the general expression (1.5) for the elastic potential of a medium with cracks; it contains vectors $\langle \mathbf{b} \rangle$ that depend on elastic constants. According to SCS, these constants are to be taken as the *effective* ones.

2-D case. Changing E_o' in (2.4) to the effective modulus E', one obtains

$$f(\sigma) = f_0(\sigma) + \frac{1}{2}(\pi/E')\rho\ \sigma_{ij}\sigma_{ij} = $$
$$= \left(\frac{1+\nu_o}{2E_o'} + \frac{\pi\rho}{2E'}\right)\sigma_{ij}\sigma_{ij} - \frac{\nu_o}{2E_o'}(\sigma_{kk})^2 \qquad (8.1)$$

(instead of 2.12 for non-interacting cracks). Identifying this expression with the potential (1.6) where E_o', ν_o are changed to the effective E', ν, one obtains an equation for the effective constants. It yields:

$$E = E_o'(1-\pi\rho)\ ; \quad \nu = \nu_o(1-\pi\rho) \qquad (8.2)$$

At small ρ, these formulas asymptotically coincide with the ones for non-interacting cracks. At larger crack densities they predict a much "softer" response. At crack density $\rho=1/\pi$ the moduli become zeros ("cut-off" point).

3-D case. Changing E_o, ν_o to the effective E, ν in (2.23) and identifying thus modified potential with (1.6) where E_o, ν_o are changed to E, ν yields two simultaneous algebraic equations for $E(\rho)$ and $\nu(\rho)$, to be solved numerically. The effective Young's modulus drops almost linearly with increasing ρ, to a "cut-off" point at ρ=9/16.

Simplified self-consistent scheme in 3-D. As discussed above, the fourth rank tensor $(1/V)\Sigma(l^3\mathbf{nnnn})^i$ in the potential (2.19) plays a relatively minor role, as compared to the α-term, and its neglection leads to relatively small errors (that can be further reduced in the formulation 2.21).

One can suggest, therefore, a simplified SCS, based on potential (2.19) with the mentioned term omitted (or, for a somewhat improved accuracy, based on potential 2.21). This leads to explicit (rather than numerically solved) expressions for the effective moduli $E(\rho)$, $\nu(\rho)$. The results are qualitatively similar to the ones of the main version (the moduli have the same cut-off point at ρ = 9/16); quantitatively, the results are very close for $E(\rho)$ and exhibit some variation for $\nu(\rho)$.

Comments on the self-consistent scheme. The scheme appears to have several drawbacks.

1. Since a representative crack is inserted into the effective matrix, *the type (and orientation) of the effective anisotropy should be specified `a priori*. The type of the overall anisotropy may be obvious for simplest orientational distribution (random orientations, parallel cracks), but it is not obvious in more complex cases.

Even if such an `a priori judgement on anisotropy *can* be made (for example, by making use of orthotropy coaxial with α), the actual realization of the scheme would be difficult, since solution for a singe crack arbitrarily oriented in an anisotropic medium would be needed - such solutions can be produced only in the numerical form. An important exception is the case of an *orthotropic* matrix containing cracks *parallel* to the orthotropy axes; in this case the calculations of the self-consistent scheme *can* be done (since the type of the overall anisotropy is obvious and a closed form solution for one crack aligned with the orthotropy axes is known), see Gottesman et al (1980).

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2. Since the impact of interactions on a given crack is simulated by inserting a crack into the material of reduced stiffness, the interactions are predicted to *always* produce a *softening* effect on the overall moduli - the curves $E(\rho)$ and $G(\rho)$ are located *below* the ones for *non-interacting* cracks.

The scheme predicts that the softening effect of interactions is *quite strong* (substantially stronger than in other approximate schemes), resulting in a "cut-off" point for the moduli in the case of random orientations. Even for *parallel* cracks (for which there is no "cut-off") the moduli rapidly drop to negligible values as ρ increases: in *3-D*, the Young's modulus drops to $0.1E_o$ at ρ=0.3 (Hoenig, 1979).

As discussed in sections 7 and 9, the actual impact of interactions on the overall stiffness may be quite different.

3. For randomly oriented cracks, the scheme predicts a "cut-off" point for the effective stiffness, at $\rho=1/\pi$ in the *2-D* case and at ρ=9/16 in *3-D*. Whether this is an advantage of the scheme (since it may be interpreted as the overall loss of integrity at high crack densities) or not, has been a subject of some discussion (Bruner, 1976, O'Connell and Budiansky, 1976). In our opinion, the scheme that reduces analysis to a *single* crack (albeit inserted into the effective medium) cannot be expected to predict the loss of integrity - phenomenon caused by crack *intersections*; the cut-off point is simply a consequence of the structure of potential (8.1) and seems to have no consistent intepretation in the framework of the scheme. Moreover, the cut-off predictions do not seem consistent: if the "loss of integrity" interpretation is accepted, the cut-off *cannot* be expected in the case of parallel cracks, as indeed shown by Hoenig; on the other hand, for the cylindrical orientation distribution (crack normals lie in parallel planes) the cut-off *should* be expected, but Hoenig's calculations *do not* predict it.

Budiansky and O'Connell (1976) suggested, as a possibility, that the actual effective stiffness may become very small near the predicted cut-off point. Computer experiments on *2-D* arrays show, however, that the Young's modulus still retains about 1/2 of its value for an uncracked material at ρ=0.3 (close to the cut-off point of $1/\pi$).

4. As remarked above, as crack density increases and interactions become stronger, the *mutual positions* of cracks become increasingly important, if the crack statistics is "biased" towards shielding or amplifying arrangements. (By translating cracks parallel to themselves, the impact of interactions may be changed from softening to stiffening, as in the example of a doubly periodic rectangular array). It is difficult to incorporate this information into the scheme.

Differential scheme (DS). Similarly to SCS, this scheme (reviewed in detail by Hashin, 1988) reduces the analysis to consideration of one isolated crack in the effective matrix. Unlike SCS, the analysis is done *incrementally*: crack density is increased by small steps $d\rho$, and the effective matrix' moduli are recalculated at each step. The procedure results in differential equations for the effective constants as functions of ρ, with initial conditions stating that $\mathbf{M}=\mathbf{M}^0$ at ρ=0. In the case of random orientations, two equations for $E(\rho)$ and $\nu(\rho)$ are coupled in the *3-D* case and uncoupled in *2-D*. Below, we consider the *2-D* case, as an illustration.

2-D, random orientations. Changing E_o' in (2.13) to the "current" E' and ρ to $d\rho$ and linearizing with respect to (small) $d\rho$ results in the differential equation for $E'=E'(\rho)$:

$$dE'/d\rho = -\pi\rho, \quad \text{with initial condition } E'(0)=E_o' \tag{8.3}$$

which yields:

$$E' = E_o' e^{-\pi\rho} \quad \text{(and, similarly, } \nu = \nu_0 e^{-\pi\rho}\text{)} \tag{8.4}$$

Unlike SCS, there is no "cut-off" point at which the stiffness vanishes.

2-D, parallel cracks. Proceeding similarly to the previous case (using 2.16 as a starting point), we obtain the Young's modulus in the direction normal to cracks:

$$E' = E_o' e^{-2\pi\rho} \tag{8.5}$$

In the case of *parallel* cracks the predicted stiffness drops much *faster* with increasing ρ than in the case of *random* orientations. At small ρ, this prediction is correct, since parallel cracks reduce the stiffness more efficiently, but at larger ρ it seems physically unreasonable, since parallel cracks experience increasing nutual shielding as ρ increases.

Comments on the differential scheme. Since DS represents an incremental version of SCS, its drawbacks are somewhat similar to the ones of SCS:

(1) *in the case of non-random orientations* and anisotropic overall properties, realization of the scheme is difficult, since it would require knowledge of a solution for one crack arbitrarily oriented in an anisotropic solid (with the exception of orthotropic matrix with cracks aligned with the orthotropy axes, see Hashin, 1988);

(2) similarly to SCS, the interactions are predicted to *always* produce the *softening* effect on the overall moduli;

(3) in the case of crack arrays "biased" towards shielding or amplification, it is difficult to incorporate an information on mutual locations of cracks into the scheme.

The softening effect predicted by DS is, generally, substantially weaker than the one predicted by SCS, and curves for the effective moduli as functions of ρ fall in-between predictions of SCS and the approximation of non-interacting cracks.

DS does not predict a "cut-off" point, and thus avoids the problem of its interpretation.

On the other hand, the results of DS *are not path-independent*: they may depend on the *order* of adding cracks of different orientations. This drawback was first noted by O'Connell and Budiansky (1976).

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We add that the simplest way to see path-dependence is to consider the case of two families of parallel cracks, each widely spaced, and forming an acute angle with each other; cracks are widely spaced so that the approximation of non-interacting cracks applies. Then the effective properties are rigorously orthotropic (see section 2). If, however, DS is applied and cracks of one family are introduced first, the new reference matrix becomes transversely isotropic; when cracks of the second family are added (at an acute angle to the plane of isotropy), the material becomes non-orthotropic - a wrong prediction. (Note that this inconsistency surfaces in quite a simple case - *2-D* array of low density).

A possible remedy for the problem of path-dependence is an imposition of the constraint that cracks of different orientations are represented in each crack density increment $d\rho$ in the same proportion as in the final configuration. The necessity of such an artificial restriction seems, however, to be a disadvantage. Another way to avoid path-dependence suggested by Sayers and Kachanov (1991) is outlined below.

A technique for extension of the self-consistent, differential (and other) schemes to arbitrary crack orientations statistics. The mentioned schemes are difficult to apply to an arbitrary orientational distribution of cracks, since this would require knowledge of a solution for a single crack arbitrarily oriented in an anisotropic solid.

Sayers and Kachanov (1991) suggested a simple technique of bypassing this difficulty (and eliminating path-dependence in DS) by using, as an input, the available results for randomly oriented and parallel cracks and tensorially transforming them to the arbitrary orientational distribution, through the crack density tensor α. Elastic potential is expressed in terms of simultaneous invariants of α and σ, with the coefficients at invariants determined from the mentioned available results. In addition to avoiding the above mentioned difficulty, this technique is in agreement with orthotropy of cracked solids.

This technique automatically transfers some of the functional dependencies of the effective moduli on ρ from one orientation distribution into another. Such a *transfer feature* may be seen as a drawback of the scheme; on the other hand, it may be used to correct the undesirable features of the approximate models.

Below, we illustrate this technique in the *2-D* case.

Recalling the expression (2.4) for the elastic potential Δf of a solid with non-interacting cracks, we modify it by assuming that the coefficient at the simultaneous invariant $(\sigma\cdot\sigma):\alpha$ is not a constant π/E_o, but a certain function of crack density:

$$\Delta f = g(\rho)\ (\sigma\cdot\sigma) : \alpha \tag{8.6}$$

where $g(\rho)$ is to be determined from the known results for either randomly oriented or parallel cracks. We then apply this expression to the case of *arbitrary* orientation statistics.

As an example, we consider DS, for which the Young's modulus for *randomly* oriented cracks $E=E_o\, e^{-\pi\rho}$ is used as an input. This implies that

$$\Delta f = (\pi e^{\pi\rho}/E_o)\ (\sigma\cdot\sigma) : \alpha \tag{8.7}$$

Applying, as a test, potential (8.7) to the case of *parallel* cracks ($\alpha = \rho \mathbf{e}_1\mathbf{e}_1$) we obtain that Young's modulus in the direction normal to cracks is:

$$E/E_o = (1+ 2\pi\rho\ e^{\pi\rho})^{-1} \tag{8.8}$$

as compared to the *actual* prediction of DS

$$E/E_o = e^{-2\pi\rho} \tag{8.9}$$

The difference between predictions (8.8) and (8.9) (disappearing in the limit $\rho \to 0$) is moderate: at $\rho = 0.3$, it is 0.17 vs 0.15; at $\rho = 0.4$, it is 0.10 vs 0.08. At larger densities, the ratio of these two values increases, but this is unimportant, since E becomes very small. Moreover, it is not clear which prediction is actually better, since at large ρ cracks experience increased mutual shielding and (8.9) may underestimate stiffness (as indeed indicated by the computer experiments of section 9); in this sense, the "transfer" feature turns out to be an advantage - it improves the scheme.

Results of DS for any other orientational distribution are obtained by simply plugging the corresponding α into the potential (8.7). Two inconsistencies of the differential scheme - path dependence and failure to predict orthotropy (which, as indicated by the data of section 9, holds for *interacting* cracks as well) - are eliminated.

In *3-D*, the technique is similar, and becomes fully identical to the one for the *2-D* case in a simplified version where the fourth rank tensor's contribution is neglected (2.21, or 2.19 with the the second term omitted).

The version presented above is somewhat simpler than the one of Sayers and Kachanov (1991), due to the fact that the term $(\sigma:\alpha)tr\sigma$ does not enter the potential (it would produce an undesirable effect on the Poisson's ratios, see comment after formula 2.10; in the work of Sayers and Kachanov the coefficient at this term is very small and their actual results are close to the ones presented here).

Being applied to SCS, this technique has the following feature: if the results for *randomly* oriented cracks are used as an input, then the "cut-off" point will be transferred to *any other* orientational distribution; if, on the other hand, the results for *parallel* cracks are used as an input, then "cut-off" is eliminated for *any* orientational distribution. This may be interpreted as an inconsistency; on the other hand, since the "cut-off" point itself may be viewed as an inconsistency, the mentioned feature may be used to eliminate it, by using the results for *parallel* cracks as an input.

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Method of Mori-Tanaka and method of effective field. Method of Mori and Tanaka (1973), often used in the mechanics of composite materials (see review of Christensen, 1990), was applied to a *2-D* cracked solid by Benveniste (1986). The analysis is reduced to consideration of *one* isolated crack placed into the *undamaged* matrix but subjected to a certain *effective* (stress or strain) *field*, which, `a priori, does not necessarily coincide with the remotely applied one.

For *cracked* solids, this method was actually developed earlier, for the *3-D* case, and in a more general form, by Kanaun (1974, 1980, 1983) who called it the "method of effective field" (and applied it to composite materials in a series of papers with Levin, see Kanaun,1983 for references). This scheme is more general, in allowing the effective field to be inhomogeneous and, thus, incorporating some information on mutual positions of cracks (for periodic configurations), although this requires extensive calculations

Predictions of this method (at least, in the simpler version when the effective field is homogeneous, i.e. the same for all cracks), in both *2-D* and *3-D*, *coincide with the ones of the approximation of non-interacting cracks* (unless additional constraints on the statistics of crack centers are imposed, for example, the requirement that circles drawn around individual cracks do not intersect, Kanaun, 1980). This degeneracy is in contrast with non-trivial results produced by this method for material with *inclusions.*

The underlying reason for the degeneracy is actually quite interesting: it seems to be a direct consequence of the fact that *introduction of cracks does not change the average stress (strain)* in a solid, provided the boundary conditions are in stresses (strains). Therefore, the seemingly degenerate results are actually correct: *there is no reason to expect the effective field* (at least in the version when it is homogeneous) *to be different from the volume average one.*

As a further remark on the method of effective field, we note that Kanaun (1980) also found the moduli in the case when crack locations are subject to a restriction: crack centers are not allowed to be inside the circles drawn around the other cracks (it is a sufficient, although not necessary, condition ensuring that crack intersections are avoided). Imposition of such a restriction makes the effective moduli slightly "softer". This "softening" seems related to creating a "bias" towards stress-amplifying configurations: the mentioned restriction appears to exclude configurations of the shielding type (parallel cracks with significant overlap) to a larger extent than the ones of the amplifying type.

Generalized ("three phase") self-consistent scheme. The generalized SCS was developed by Christensen and Lo (1979) for composite materials. Being applied to a *2-D* cracked solid (Aboudi and Benveniste, 1987), it assumes that the crack is, first, inserted into a circular inclusion of compliance $\mathbf{M}^0$ (of the undamaged matrix) and, second, this cracked inclusion is inserted into the material having *effective* compliance $\mathbf{M}$.

Such modelling attempts to account for the fact that the immediate vicinity of a crack is stiffer than the homogenized effective material. Predictably, this scheme produces effective moduli that are stiffer than the ones of SCS. The results of the generalized SCS are quite close to the ones of DS.

One of the difficulties encountered in the framework of this model is the problem of *overlapping* of the circular inclusions containing cracks. This problem sets limitations to the highest crack density that can be considered (about ρ=0.3); it also interferes with the assumption of *randomly located* cracks. Also, it seems that the calculations of this model would be difficult in the *3-D* case.

Models of the second order in crack density. Two such models have been proposed: the one of Hudson (1980, 1981, 1986) and the one of Ju and Chen (1992); they calculate effective stiffnesses and effective compliances, correspondingly, that are correct to the second order in crack density, in the cases of randomly oriented and aligned cracks.

The results of Hudson were plotted by Sayers and Kachanov (1991); as seen from these plots, at low crack densities the results follow predictions of DS; at moderate-to- high densities (of the order of $\rho = 0.2$ for parallel cracks and $\rho = 0.6$ for randomly oriented cracks) they start to exhibit a physically meaningless behaviour: the stiffness *increases* with increasing crack density.

The results of Ju and Chen (for randomly oriented and parallel cracks) based on ensemble averaging, are free of this inconsistency. However, it seems that, in the case of randomly oriented cracks, the authors do not properly address the (difficult) problem of avoiding crack intersections, and it is unclear whether this inaccuracy introduces errors of the same ρ^2-order or not. Note that, in the case of parallel cracks, their coefficient at the ρ^2-term seems to be at variance with calculations of Horii and Sahasakmontri (1990); the underlying reason for the difference is not fully clear to the present author.

As a general remark on the second order terms, (assuming they *are* properly constructed), we note that the interval of ρ over which the ρ^2-term actually provides improvement (combined effect of higher order terms is smaller than the effect of the ρ^2-term) remains to be investigated.

Model for parallel arrays of very high density. Wu and Chudnovsky (1990) considered an array of parallel cracks in the limit of very high density. They suggested a beam model for calculation of the Young's modulus in the direction normal to cracks, and found upper and lower bounds, based on clamped beam and simply supported beam assumptions, correspondingly.

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Concluding remarks. The self-consistent, the differential and Mori-Tanaka's schemes reduce the interaction problem to the problem of *one* crack placed into effective homogeneous environment (with an intermediate stiffer "cushion" in the generalized self-consistent scheme). With the exception of Mori-Tanaka's scheme, they predict that interactions produce a *softening* effect on the effective moduli; the strongest softening is predicted by the self-consistent scheme. None of these schemes is capable of incorporating an information on mutual crack positions.

Mori-Tanaka's scheme (originally developed for composites) and Kanaun's method of effective field degenerate for cracked solids - the predicted moduli coincide with the ones for *non-interacting* cracks. This is not an incostistency, but properly reflects the actual situation: since introduction of cracks does not change the average stress in the matrix (provided *tractions* are prescribed at boundaries), there is no reason to expect any impact of interactions on the effective moduli if crack locations are random (see fig. 6).

Kanaun's method of effective field allows the effective field to be inhomogeneous (different for different cracks) and, thus, may take the information on mutual crack positions into account (as demonstrated by application to some periodic arrangements); in this more general version, however, the calculations become lengthy.

Models that are second order in crack density may be promising (since they may incorporate, in some approximate way, an information on mutual crack locations), provided that, firstly, the problem of avoidance of intersections is properly addressed (so that the second order term is indeed exact), and, secondly, an estimate is given assuring that the total effect of the neglected higher order terms is small as compared to the one of the second order term.

Curves for various approximate schemes, for randomly oriented and parallel cracks in isotropic and orthotropic matrices, in *2-D* and *3-D*, were plotted and compared by several authors; see, for example Kemeny and Cook (1986), Hashin (1988), Sayers and Kachanov (1991).

Zimmerman (1985) made an interesting observation that the relation *between* different moduli (say, between the Young's modulus E and the shear modulus G) is almost independent of the scheme used, so that it is difficult to verify the approximate schemes by experimentally measuring the E-G curve.

The approximate schemes aim at improving the results of the approximation of the non-interacting cracks. As discussed in section 9, this is not necessarily achieved.

9. SOLUTIONS FOR DETERMINISTIC ARRAYS. COMPUTER EXPERIMENTS.

Method of solution. Solutions for *deterministic* arrays of arbitrary geometry, in both *2-D* and *3-D*, can be produced, by relatively simple means and with good accuracy, using the method of analysis of many cracks' problems developed by Kachanov (1987). This method is briefly outlined below in *2-D* version (extension to *3-D* is straightforward, see Kachanov and Laures, 1989).

The problem of a linear elastic solid with N cracks subjected to stress σ at infinity is equivalent to the problem with crack faces loaded by tractions $\mathbf{n}^i \cdot \sigma$ ($i = 1,...,N$) and stress free boundary. The latter problem can be represented as a superposition of N sub-problems, each containing only *one* crack, but loaded by *unknown* tractions, comprising, in addition to the σ- induced traction, extra tractions accounting for interactions that have to be found. In *2-D* problems, such superpositions (usually combined with polynomial expansions of tractions and interrelating the polynomials' coefficients through a large system of linear algebraic equations) have been used starting from the early 70's (see Kachanov,1987 for references); in *3-D* problems, they were used by Kachanov and Laures(1989).

Thus, traction $\mathbf{t}^i$ in the i-th sub-problem is a sum of the σ -induced traction $\mathbf{n}^i \cdot \sigma$ and tractions induced by the other cracks in the remaining sub-problems along the line l^i:

$$\mathbf{t}^i(\xi^i) = \mathbf{n}^i \cdot [\sigma + \sum_{j \neq i} \sigma^j(\xi^i)] \equiv \mathbf{t}^{i\,(o)} + \Delta\mathbf{t}^i(\xi^i) \quad (9.1)$$

where ξ^i is a current point on l^i and σ^j is the j-th crack-generated stress (fig.6).

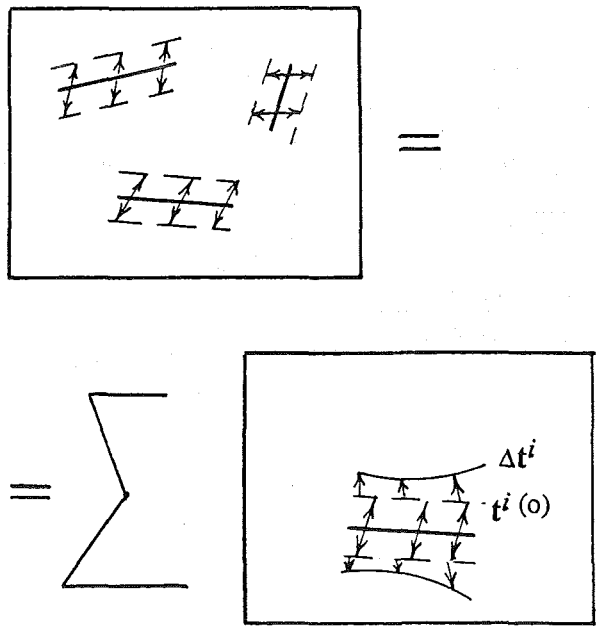


FIG.6. Stress superpositions. Traction $\Delta\mathbf{t}^i(\xi^i)$, induced on the i-th crack by the other cracks, accounts for interactions; it may either increase or reduce $\mathbf{t}^i$. If crack locations are random (and *tractions* are prescribed on the boundary), the average of $\langle\Delta\mathbf{t}^i\rangle$ over the array is zero - there is no impact of interactions on the effective moduli.

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The basic simplifying assumption of our method is that σ^j is taken as generated by a *uniform average* traction $\langle \mathbf{t}^j \rangle$, yet unknown; the impact on the i-th crack of traction non-uniformities $\mathbf{t}^j - \langle \mathbf{t}^j \rangle$ along the j-th crack is neglected. Decomposing the average $\langle \mathbf{t}^j \rangle$ into a sum of normal and tangential averages, $\langle p^j \rangle$ and $\langle \tau^j \rangle$, we have

$$\mathbf{t}^i(\xi^i) = \mathbf{n}^i \cdot \sigma + \mathbf{n}^i \cdot \sum_{j \neq i} [\langle p^j \rangle \sigma_n^j(\xi^i) + \langle \tau^j \rangle \sigma_\tau^j(\xi^i)] \quad (9.2)$$

where σ_n^j and σ_τ^j are the "standard" stress fields, generated (in an infinite solid) by a single j-th crack loaded by uniform tractions of unit intensity, normal and shear, correspondingly. These fields (expressed in elementary functions) are available in fracture mechanics literature.

The unknown quantities in (9.2) are the average tractions $\langle \mathbf{t}^j \rangle$ on cracks. They are found by averaging (9.2) along the i-th crack line and thus interrelating them by a system of N vectorial ($2N$ scalar) linear algebraic equations

$$\langle \mathbf{t}^i \rangle = \mathbf{n}^i \cdot \sigma + \sum_{j=1}^{N} (\Lambda^{(ji)} - \delta^{ji}\, \mathbf{I}) \cdot \langle \mathbf{t}^j \rangle \quad (9.3)$$

where δ^{ji} is Kronecker's delta and the (tensorial) transmission Λ - factors characterize transmission of the average tractions from one crack onto another ($\Lambda^{(ji)} \cdot \langle \mathbf{t}^j \rangle$ is the average traction induced on a line l^i in a continuous material by the j-th crack loaded by a uniform traction $\langle \mathbf{t}^j \rangle$). The Λ - factors are expressed in terms of line integrals of elementary functions and are easily calculated.

Solving (9.3) for average tractions, one obtains:

$$\langle \mathbf{t}^i \rangle = \sum_{j=1}^{N} \Omega^{(ji)} \cdot \langle \mathbf{t}^j \rangle = \sum_{j=1}^{N} \Omega^{(ji)}\, \mathbf{n}^j : \sigma \quad (9.4)$$

where $\Omega^{(ji)} = (2\delta^{ji}\, \mathbf{I} - \Lambda^{(ji)})^{-1}$. The case of non-interacting cracks is recovered by assuming $\Lambda^{(ji)} = \delta^{ji}\, \mathbf{I}$; then $\mathbf{t}^i = \mathbf{n}^i \cdot \sigma$.

Finding the average displacement discontinuity $\langle \mathbf{b}^i \rangle$ as approximately proportional to the average traction $\langle \mathbf{t}^i \rangle$ with the proportionality coefficient $\pi l^i / E_o'$ (Kachanov, 1987) we obtain *an explicit expression for the effective compliance*:

$$\mathbf{M} = \mathbf{M}^0 + (2\pi / A\, E_o') \sum_{i,j=1}^{N} l^{(i)2}\, \Omega_{\{}^{(ij)}\, \mathbf{n}^i\, \mathbf{n}^j{}_{\}} \quad (9.5)$$

(subscript braces denote all the appropriate symmetrizations, i.e. with respect to $p \leftrightarrow q$, $r \leftrightarrow s$, and $pq \leftrightarrow rs$ for M_{pqrs}). It incorporates information not only on the crack orientations and sizes, but, also, on their *mutual positions* (all these factors are reflected in Λ's, and, thus, in Ω's).

The method yields results for *given arrangements* of cracks; they are not readily translated into statistical terms. On the other hand, such results constitute direct computer experiments on sample arrays and can be used to verify various approximate schemes. The method is accurate up to quite close spacings between cracks so that the results are accurate up to high crack densities (as confirmed by smallness of corrections provided by the alternating technique).

As noted above, there is no reason to expect the results to be different from the ones predicted by the approximation of non-interacting cracks (provided crack locations are random). This conclusion was directly verified by computer experiments on sample arrays.

Computer experiments. Sample *2-D* crack arrays contained 25 cracks and were generated with the help of random number generator, as realizations of certain crack statistics.

Two *orientation statistics* were assumed: randomly oriented cracks and parallel cracks; for each of them, six crack densities were assumed: $\rho = 0.10$; 0.15; 0.20; 0.25; 0.30 and 0.35 (in *2-D*, the densities of 0.25-0.35 can be considered as quite high).

Fifteen sample arrays were considered for each density.

Locations of cracks within the representative area were random, i.e. positions of crack centers were uncorrelated. For *parallel* cracks, generation of such arrays was straightforward. For *randomly* oriented cracks, crack intersections had to be avoided; this was achieved by generating cracks successively and discarding the newly generated one if it intersects the already existing cracks and generating it again. Although such procedure, strictly speaking, violates the condition that crack centers are uncorrelated, we assume that it does not create errors of a *systematic* sign.

The sample arrays usually contained several cracks with spacings between them much smaller than the crack sizes; therefore, the method outlined above was supplemented by the *alternating technique* (stress "feedbacks") for improved accuracy, until the three digits accuracy for the effective moduli was reached. It actually produced only an insignificant correction (smaller than 3-4%) of the method's results, confirming accuracy of the method.

In order to limit the number of iterations in the alternating technique, the spacings between cracks were not allowed to be overly small. Namely, in the case of randomly oriented cracks, they were not smaller than 0.02 of the crack length; in the case of parallel cracks, they were not smaller than 0.15 of the crack length for those crack pairs that had a significant "overlap", and not smaller than 0.02 of the crack length otherwise.

In the case of randomly oriented cracks, each array was examined for *isotropy*: the effective Young's modulus was calculated in two perpendicular directions and only those arrays were kept for which the difference between these two

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moduli was smaller than 2%; the Young's modulus was then taken as the average over these two values. (Note that preferential orientations causing up to 5-7% variation of E with direction are usually not discernible by a "naked eye").

Fig.7 shows some of the generated arrays.

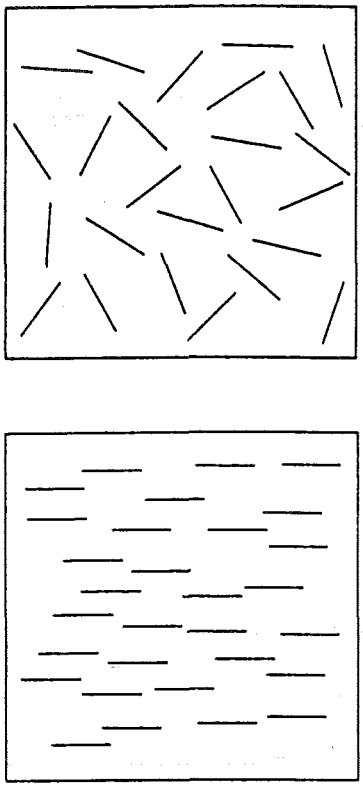


FIG.7. Two of the crack arrays examined (randomly oriented, $\rho = 0.25$ and parallel, $\rho = 0.30$).

As discussed in Introduction, the representative area A is assumed to constitute a part of a statistically homogeneous field of cracks (extending all the way to the external boundary) and stresses at the boundary Γ of A are assumed constant and equal to the remotely applied ones. This assumption is rigorously correct for non-interacting cracks, when cracks inside A do not experience any influence of those outside A. For *interacting* cracks, stresses fluctuate along Γ. The errors due to the mentioned assumption depend on the size of the sample. They cause *scatter* of results from one realization of the crack field statistics to another, but are not expected to produce errors of a *systematic* sign. Figs. 8 and 9 show that the scatter is small enough for the results to definitely indicate a certain trend.

The effective moduli for each array were first calculated in the assumption of non-interacting cracks; the results coincided with the theoretically predicted values to within three digits (minor differences were due to an imperfect isotropy). Then the procedure for computing the interaction effect was "turned on" and the moduli calculated again.

Results. Figs.8 and 9 show results for the Young's modulus for randomly oriented and parallel arrays. Vertical bars show scatter of the results from one sample to another.

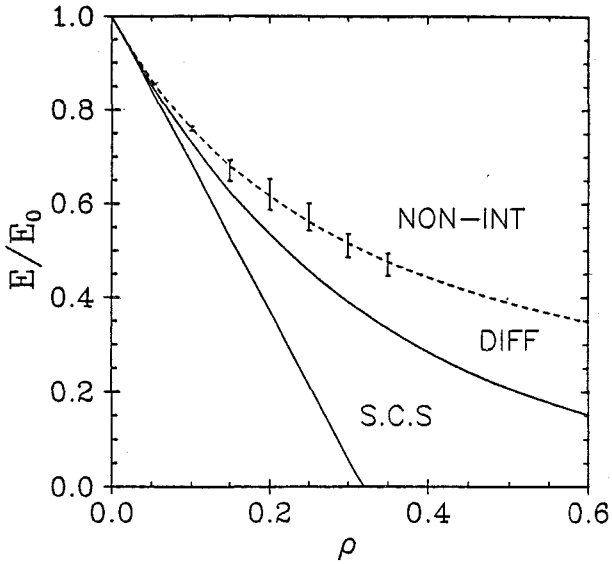


FIG.8. Effective Young's modulus, randomly oriented cracks: results for sample arrays vs predictions of the approximation of non-interacting cracks, self-consistent and differential schemes.

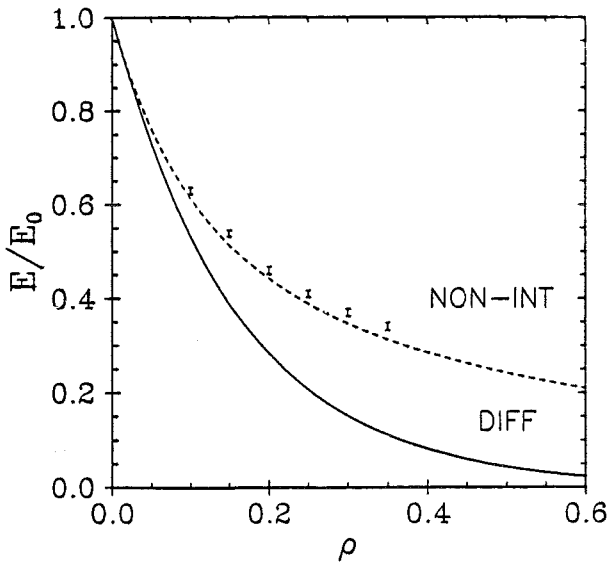


FIG.9. Effective Young's modulus, parallel cracks: results for sample arrays vs the approximation of non-interacting cracks and the differential scheme.

For *randomly oriented* cracks, the approximation of non-interacting cracks provides surprisingly good results, well into the domain of strong interactions where this approximation is usually considered inapplicable.

For *parallel* cracks, the approximation of non-interacting cracks still provides good results. However, there is a slight but distinguishable tendency for the stiffening overall effect of interactions, indicating a slight dominance of the shielding mode of interactions. This tendency is in

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contradiction with the result of Willis (1980) who proved that, in the case of parallel cracks with randomly located centers ("well stirred" cracks, in his terminology) the approximation of non-interacting cracks provides the upper bound for the effective Young's modulus (in the sense of bounding the *ensemble average*; bounding applicable to *any* realization of statistics is impossible, see section 10). The reason for this contradiction is not fully clear to the present author. We suspect that the prohibition for spacings between substantially overlapping cracks to be smaller than 0.15 of the crack length (that made the sample arrays slightly non-random) may have created a slight "bias" in favour of shielding: since the amplification zone extends over smaller distances in the direction normal to cracks than the shielding zone (fig.5), elimination of closely spaced overlapping cracks may have reduced the effects of amplification to a larger extent than the ones of stress shielding.

We also calculated the shear modulus for parallel crack arrays (for two arrays, $\rho = 0.15$ and 0.20). It differed from the predictions for non-interacting cracks by less than 1.5%.

Note that the *stress intensity factors* (SIFs) at many crack tips were substantially affected by interactions, thus indicating that interactions in the sample arrays were strong.

Preservation of orthotropy for interacting cracks. As discussed in section 2, for *non-interacting* cracks orthotropy (coaxial to α) is a rigorous result in *2-D*.

Consistently with the fact that, due to cancellation of competing effects, the approximation of non-interacting cracks remains accurate at high crack densities, *orthotropy* can also be expected to hold at high ρ.

We examined *deviation from orthotropy* in sample arrays of non-randomly oriented (but randomly *located*) cracks, by solving the interaction problem for two families of parallel cracks of equal density inclined at 30^0 to each other. At $\rho = 0.24$ (significant interactions, as indicated by substantially affected SIFs at many crack tips) the deviation from orthotropy produced by interactions was very small: in the principal coordinate system of α the non-orthotropic compliances ΔM_{1112} and ΔM_{2212} were 10^{-3}- 10^{-4} of the orthotropic compliances, i.e. indistinguishable from zero (within the accuracy of the results).

This means that *characterization of crack arrays by the crack density tensor remains adequate even at high crack densities* when interactions are strong, and implies that the "natural" coordinate system (coaxial to α) remains the system of choice for calculation of the effective moduli.

Discussion. The results show that, well into the interval of crack densities (ρ up to 0.35) where the interactions become significant (as indicated by significantly affected SIFs), *the approximation of non-interacting cracks remains quite accurate.*

The underlying reason for this accuracy is *not* that the interactions can be neglected, but that the competing effects of stress shielding and stress amplification *cancel* each other.

This is a direct consequence of the fact that introduction of traction-free cracks does not change the average stress (provided *tractions* are prescribed on the boundary). Therefore, if a certain number of "new" cracks is introduced into the environment of the preexisting ones in a *random fashion*, these new cracks will, *on the average*, experience no effect of the preexisting cracks. In the language of superpositions (fig.6), the additional tractions $\Delta \mathbf{t}^i$ will be of both amplifying and shielding nature on different cracks; on the average, their impacts will cancel each other.

We emphasize that this conclusion assumes the *absence of "bias"* in crack statistics towards either amplifying or shielding arrangements. Otherwise (in "ordered" arrays, for example) the impact of interactions can be very large, both in the direction of "softening" and "stiffening".

The reported results are for the *2-D* configurations. The approximation of non-interacting cracks will remain accurate in *3-D* as well, due to the same mechanism of cancellation of shielding and amplification effects. Moreover, we expect the scatter of the results to be smaller, since interactions are much weaker in *3-D* (Kachanov and Laures, 1989).

The differential scheme and, in particular, the self-consistent scheme substantially overestimate the impact of cracks on the effective stiffness. The underlying reason is that the impact of interactions on a given crack is simulated by a *reduced stiffness of the surrounding material*; therefore, interactions are predicted to *always* reduce stiffness. Such modelling does not take into account the actual mechanisms of interaction (see superpositions of fig.6). In particular, the account is not taken of the shielding mode of interactions that produces a *stiffening* effect and neutralizes the impact of the amplifying mode.

On the other hand, our results are in agreement with Mori-Tanaka's scheme (and the method of effective field, at least, in the version of homogeneous field) which correctly predicts that the effective field acting on a representative crack is the same as in the absense of interactions.

Concluding remark. The conclusion on accuracy of the approximation of non-interacting cracks is based on the assumption of *randomness of crack locations*.

Such randomness may not always be the case in *naturally occurring* crack systems. A crack may, possibly, enhance nucleation of new cracks in its amplification zone and suppress it in the shielding zone, thus creating an amplification "bias". This may be relevant for the stages of microcracking when localization is about to occur. As an opposite example, the observations of Han and Suresh (1989) on microcracking in ceramics seem to indicate (although not fully conclusively) some shielding "bias" in most (but not all) microcracking geometries (Suresh and Kachanov, to be published). In such cases, the results of this section should be used with caution.

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10. ON BOUNDING THE EFFECTIVE MODULI.

Bounding of the moduli, widely used in the mechanics of two components materials, *degenerates* for cracked solids: neither lower nor upper bounds (other than trivial) can be constructed, at least, in the sense of bounding valid for *any particular* sample of a given crack statistics (as versus bounding for the ensemble average).

We note that the opposite statement - that the approximation of non-interacting cracks provides the upper bound for *any* crack statistics - was made by Gottesman et al (1980). Such a general statement (implying that interactions *always* reduce stiffness) seems questionable - in *shielding* arrangements (stacked cracks, for example), interactions produce the *opposite* effect of toughening.

Willis (1980) proved that, for *parallel* cracks with uncorrelated positions of crack centers, the approximation of non-interacting cracks constitutes the upper bound in the more *restricted* sense of bounding the *ensemble average*.

To demonstrate the impossibility of bounds that are valid for *any* realization of a given orientational distribution, we construct two types of configurations:

(A) the one with crack density $\rho \to \infty$ that produces a vanishingly small effect, and

(B) the one with $\rho \to 0$ that has, nevertheless, a vanishingly small stiffness.

We restrict attention to *2-D* geometries and the cases of isotropic and transversely isotropic effective properties.

In the case of *isotropic* effective properties, we establish a fully symmetric hexagonal mesh (H is a side of a cell) and consider the following two configurations.

(A) Radial crack patterns contained within circles of radius r are placed at vertices of the mesh. Making radial patterns arbitrarily dense, while keeping the circles radii sufficiently small ($r << H$) results in a configuration where an *arbitrarily large crack density ρ produces a vanishingly small impact* on the effective moduli (fig.10).

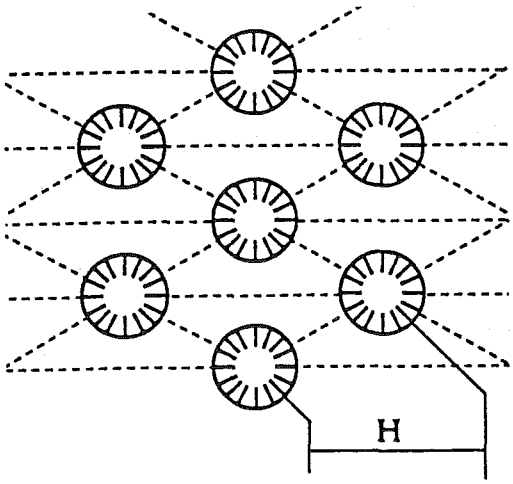


FIG.10. Crack array of infinite crack density, $\rho \to \infty$ producing an infinitesimal impact on the effective moduli.

(B) Three families of periodically spaced collinear cracks (crack length is $2l$ and ligaments are Δ) are placed along the mesh (fig.11). Both $2l$ and Δ are then made progressively smaller, in such a way that Δ decreases faster than $2l$; namely, assuming that H contains N periods: $N(2l+\Delta)=H$, we set $l = 0.5H/(N+1)$ and $\Delta=H/(N^2+N)$; then the crack density $\rho = (4\sqrt{3})^{-1}N/(N+1)^2$. We let $N\to\infty$, then the total relative length of ligaments $N\Delta/H = 1/(N+1)\to 0$ so that *the effective stiffness* $\to 0$. At the same time, $\rho \to 0$.

We further note that, since ρ and $N\Delta/H$ are independent of H, the parameter H can also be set to $\to 0$ as N increases (say, $H=H_0/N$). This creates a fine "mist" of infinitesimal cracks that is *dense everywhere* while possessing the property that $E \to 0$ and, at the same time, $\rho \to 0$.

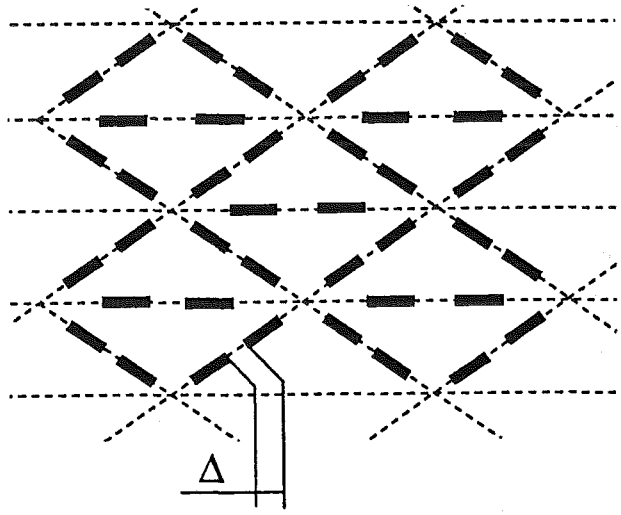


FIG.11. Crack array of infinitesimal crack density, $\rho \to 0$ that reduces the effective stiffness to zero. Note that this effect is achieved without creating a line of cracks crossing the representative area - the array is dense everywhere (since $H \to 0$).

In the case of *transversely isotropic* effective properties, configuration of the type (A) is constructed as a doubly periodic rectangular pattern of parallel cracks; spacings between different columns are kept sufficiently wide, while the density of packing within each column $\to \infty$. Configuration of the type (B) is constructed as the same pattern, with vanishingly small ligaments between collinear cracks and widely separated rows.

We emphasize that the statement on impossibility of non-trivial bounds assumes that *any possible arrangement of crack centers* is allowed. If *restrictions* are imposed on the statistics of crack centers, or if bounding is understood as holding for the *ensemble average* of the given crack statistics, then non-trivial bounding may be possible (as, for example, in the work of Willis, 1980).

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11. ALTERNATIVE CRACK DENSITY PARAMETER.

Remarks on conventional (location insensitive) crack density parameter. Both the scalar crack density ρ and its tensorial generalization - crack density tensor α are *insensitive to mutual positions* of cracks. This is fully adequate for *non-interacting* cracks, but may become inadequate for interacting cracks, in particular, when crack statistics is "biased" towards shielding or amplificatying arrangements.

This inadequacy results, for example, in *non-equivalence* of the *assumptions of non-interacting cracks and of small crack density* (see examples of section 10) - an undesirable feature. The *impossibility of bounding* in terms of ρ can also be traced to insensitivity of ρ to crack locations.

Alternative crack density parameter. An alternative parameter that is sensitive to crack positions and, in the absence of interactions, reduces to α, can be constructed as follows. We restrict our consideration to the *2-D* case; extension to *3-D* is straightforward.

We recall that the change Δf in the elastic potential due to cracks has the form:

$$\Delta f = (1/A) \sum_i (\mathbf{n}\cdot\sigma\cdot\langle\mathbf{b}\rangle)^i l^i \qquad (11.1)$$

Since $\langle \mathbf{b}^i \rangle \approx (\pi l^i / E_o')\langle \mathbf{t}^i \rangle$, we obtain, expressing $\langle \mathbf{t}^i \rangle$ from (9.4) in terms of transmission factors:

$$\Delta f = (\pi/E_o') \, \sigma : \omega : \sigma \qquad (11.2)$$

where the fourth rank tensor

$$\omega = \frac{1}{A} \sum_{i,k} l^{(k)2} \mathbf{n}^k \Omega^{(ki)} \mathbf{n}^i \qquad (11.3)$$

that emerges in the potential, accounts not only for the crack sizes and orientations but for the mutual positions as well, and *can be used as a parameter characterizing crack array.*

In the case of non-interacting cracks, $\Omega^{(ki)} = \delta^{ki}\mathbf{I}$ so that $\sigma : \omega : \sigma = \sigma\cdot\sigma : \alpha$ and Δf reduces to the expression (2.4) in terms of the crack density tensor α.

Thus, ω can be considered as a *generalization of the crack density tensor that accounts for crack interactions and is sensitive to the mutual crack positions.*

We note that smallness of ω means smallness of a sum of products (α-type terms)$\times$ (Ω-terms). The Ω– terms, as seen from their definition (9.4), characterize average tractions on cracks as fractions of the tractions *in the absence of interactions*; thus, the Ω - terms characterize shielding, if they are small, or amplification, if they are large.

Therefore, if ω (rather than ρ or α) is used as a parameter, "paradoxical" situations when infinitesimal crack density reduces the effective stiffness to zero, or an infinite density producing only an infinitesimal effect (examples of section 10) are eliminated. Indeed, for a dense radial pattern of fig.10 (or densely packed stacks of parallel cracks) the conventional density ρ may be large, but ω is *small*, since the Ω - terms are small (due to a strong shielding). On the other hand, for the hexagonal pattern of fig.11 the parameter ω is large (although $\rho \to 0$), due to a strong amplifying effect of interactions.

Thus, problems related to "bias" towards shielding or amplifying configurations are eliminated.

Another inconsistency of the theories using ρ or α - non-equivalence of the assumptions of non-interacting cracks and of small crack density - is also eliminated, since ω is small *if and only if* the impact of cracks on the effective properties is small.

Relevance of ω for randomly located cracks and for "ordered" arrays. For randomly located cracks, actual evaluation of ω presents a difficult problem associated with geometrical probabilities (inavoidable, though, if one wishes to incorporate the information on mutual crack positions).

On the other hand, the results of section 9 indicate that, for *randomly located* cracks, the approximation of non-interacting cracks yields reasonably good results (at least, up to crack densities of $\rho = 0.35$, which, in *2-D*, can be considered as relatively high), for both the values of compliances and the symmetry (orthotropy) of the effective properties. Therefore, descriptions in terms of α seem to remain adequate, and switching to ω may be unnecessary.

For "ordered" arrays that may have "bias" towards shielding or amplification, deviations from the predictions of the approximation of non-interacting cracks may be very large. Then, the use of ω, instead of α, appears to be important. At the same time, evaluation of ω for such arrays seems a more realistic task, since a number of available solutions can be utilized.

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12. FRACTURE OF MICROCRACKING SOLIDS AND CHANGE OF EFFECTIVE MODULI: ANY CORRELATION ?

It is argued that, contrary to the spirit of many damage models, there is no stable quantitative correlation between fracturing of a brittle microcracking solid and the change of its effective elastic moduli (with an obvious exception of the immediate vicinity of the fracture point, when the effective stiffness drops to zero). Therefore, the change of effective moduli may not always serve as a reliable indicator of the progression of a microcracing solid towards fracture.

Physically, the absence of such correlations is explained by the fact that the fracture-related quantities (like SIFs) are determined by *local fluctuations* of the microcrack field geometry whereas the effective moduli are *volume average* quantities, relatively insensitive to such fluctuations.

This discussion is relevant for a number of damage models (see, for example, review of Krajcinovich, 1989) aimed at description of *fracturing* of a brittle microcracking solid, but in fact dealing with its *effective elastic constants*. The underlying idea - that progression towards failure is uniquely correlated with the change of effective moduli - may seem intuitively reasonable; it appears particularly attractive because the moduli can be easily measured.

An objection may be raised that a small crack in a brittle material has a very small impact on the effective moduli but may drastically reduce the tensile strength. One may argue, however, that, after a certain initial set of cracks has been nucleated and started to grow, the quantitative correlation *can* be established. This idea is examined below from several points of view.

"Paradoxical" example. To demonstrate that this idea is far from obvious, we start with a simple example when the correlation between the SIFs and the effective moduli is "paradoxical". Consider a plate containing stacks of parallel cracks (fig.12, solid lines), and additional cracks are introduced in-between the neighbouring cracks (dashed lines).

Introduction of new cracks reduces elastic stiffness in the direction normal to cracks - material *elastically softens*. However, the SIFs *decrease* (due to increased shielding) - new cracks produce a *strengthening* effect.

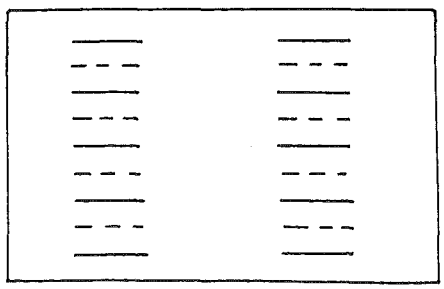


Fig.12. Introduction of new cracks (dashed lines) reduces the SIFs and, at the same time, produces elastic "softening"

Random crack arrays. It may be argued that the example above does not represent any realistic crack statistics (although such parallel crack patterns do occur in rocks and certain composites) and that, for more "random" arrays, progression towards failure *is* uniquely correlated with the reduction of elastic stiffness.

To demonstrate that such a correlation is not obvious, we did the following computer experiments. A number of two-dimensional crack arrays containing randomly oriented cracks of the same length were generated; statistics of crack centers was also random (subject to the restriction that cracks do not intersect. For each sample array, using the method described in section 9, we calculated

(1) the effective Young's modulus E_{eff};

(2) maximal, among all crack tips, value of $K_I^2+K_{II}^2$ induced by a uniaxial loading. (Strictly speaking, the value of $K_I^2+K_{II}^2$ determines the initiation of crack propagation only in the case of rectilinear crack extension; if both K_I and K_{II} are present, a certain other combination of K_I ,K_{II} is a relevant parameter. Typically, however, this combination does not differ much from $K_I^2+K_{II}^2$).

We found that, whereas E_{eff} was quite stable from the one statistical sample to another, $Max\ (K_I^2+K_{II}^2)$ *fluctuated significantly*, reaching higher values in larger samples reflecting a higher probability of occurrence of closely spaced cracks.

Thus, the stiffness E_{eff} quickly stabilizes as the size of the crack sample increases (usually, at the number of cracks of the order of 10) whereas the fracture-related parameter $Max\ (K_I^2+K_{II}^2)$ is statistically unstable, and increases, as the size of the crack sample increases.

Strong and weak interactions. Clustering of cracks. Interactions that produce*strong* impact on the SIFs may be *weak* in their impact on the effective moduli. (As discussed in section 9, in the case of randomly located cracks, the effective moduli are well predicted by the approximation of *non-interacting* cracks).

A closely related issue is that *clustering* of cracks strongly affects the SIFs. Fig.13 provides an illustration: $Max\ (K_I^2+K_{II}^2)$ is, typically, substantially higher in configurations of the type 13b than in ones of the type 13a.

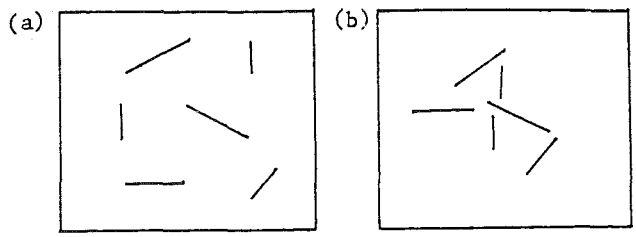


Fig.13. Clustering of cracks

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On the other hand, sensitivity of the effective elastic moduli to clustering is low, i.e. the difference in moduli of the configurations 13a and 13b is generally insignificant (provided the overall crack density is the same).

Thus, monitoring the change in effective elastic constants may not necessarily detect the onset of strong interactions and clustering of defects - events crucially important from the point of view of progression towards failure.

Effect of the crack shape. Consider a *3-D* elastic solid containing elliptical cracks. The SIFs and their maximal values along the crack edges depend on the aspect ratios of the ellipses. At the same time, the shapes of cracks have a very small impact on the overall effective elastic stiffness (see discussion after formula 2.37 of Budiansky and O'Connell, 1976).

Crack-microcrack interaction problem. Fracture propagation in brittle inhomogeneous materials is often accompanied by microcracking (see, for example, experimental observations on high temperature fracture of ceramics of Han and Suresh, 1989). Elastic interactions with microcracks may either reduce or increase the SIFs at the main crack (stress shielding or stress amplification) thus producing either toughening or enhancing effect.

In the crack-microcracks problem, the absence of correlation between the effective moduli and the fracture-related properties is apparent. As discussed by Kachanov et al (1990), microcracks located in the *short range interaction zone* (several microcracks closest to the main crack tip) produce a *dominant* impact on the main crack tip. Due to a high sensitivity of the interaction effect to the exact positions of closest microcracks, the impact of microcracks on the main crack SIFs fluctuates significantly and even qualitatively (from shielding to amplification) from one sample of the microcrack statistics to another.

This means that there is *no statistically stable effect of stress shielding*, at least, if the microcrack locations are random. On the contrary, being conservative, one may conclude that the overall offect of interactions is stress amplification, particularly in *3-D* where the *local peaks* of SIFs along the crack front are almost always present (in spite of the fact that the average along the front SIF may experience a mild shielding); these local peaks *enhance* local crack advances (Laures and Kachanov, 1991).

On the other hand, replacing the microcracked region by an "effective" material of reduced stiffness would produce a different result - that interactions result in shielding.

Another important effect is the appearance of "secondary" modes on the main crack, due to stochastic asymmetries in the microcrack field, i.e. K_{II} under mode I loading. Since K_{II} promotes crack kinking, this factor may be partially responsible for an irregular shape of crack paths in microcracking materials. This effect is, obviously, missed if the microcracked zone is replaced by an effective elastic material.

We note that the arguments above on the absence of shielding do not exclude, of course, the possibility of microcrack toughening by mechanisms *other than elastic interactions*, like expenditure of energy on nucleation of microcracks; neither does it exclude the possibility of stress shielding due to interactions with some *special* microcrack arrangements, like an extremely dense array of microcracks parallel to the main crack (Chudnovsky and Wu, 1990) or for the statistics of the microcrack centers that is not random but "biased" towards shielding configurations, as seems to be the case in some of the experimental observations of Han and Suresh, 1989).

Cases when progression towards fracture can be monitored by the change of effective moduli. The basic underlying reason for the absence of the one-to-one correspondence between progression towards fracture and change of hte effective elastic constants is that the fracture-related quantities (like SIFs) are determined by the *local fluctuations* of the microdefect field geometry whereas the effective elastic properties are the *volume average* quantities, relatively insensitive to such fluctuations.

Therefore, in the cases when evolution of the defects' population follows a more or less *deterministic reproducible pattern*, the progression towards fracture may indeed be monitored by the reduction of the effective elastic stiffness. Such situations occur in certain laminated composites where, under certain loading conditions, the formation of microcracks follows well established patterns (called "characteristic damage states" by Reifsnider and Stinchcomb, 1983). We note, however, that, even in these cases, the correlation is not straightforward: as their data shows, (a) the correlation between the fraction T of the remaining lifetime of a specimen and its effective stiffness E holds only for a *given arrangement* of layers. If the ply orientations, and, therefore, the crack patterns, are changed, the E - T curve changes; (b) one of the E - T curves remained almost *flat*, i.e. the stiffness remained almost insensitive to damage development, *during most of the lifetime*.

On models of the $f = (1-D)f_0$ type. It has been suggested by a number of authors (see Lemaitre and Chaboche, 1978, for example) that the effect of microcrack-related damage can be characterized by specifying the elastic potential in strains in the form

$$f(\varepsilon) = (1-D)f_0(\varepsilon) \qquad (12.1)$$

where D is some scalar measure of damage and f_0 is the potential without damage. (This form is, obviously, limited to *isotropic* damage - an attempt to apply it to *parallel* cracks, for example, would predict the equal stiffnesses in the directions normal and parallel to cracks).

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We note that, regardless of other applications, *any model for f necessarily implies a certain model for the effective elastic moduli*. From this point of view, the model has major inconsistencies. One of them was pointed out by Ju (1990): in the case of randomly oriented cracks ("scalar damage"), the Poisson's ratio remains constant, as damage increases - an unrealistic prediction.

We add that, in the limit of *small* D, the model is not consistent with the rigorous results for non-interacting inhomogeneities, whether these inhomogeneities are cracks or cavities and D is a usual measure of their density, rigorously emerging in the elastic potential.

Thus, even if D is understood as simply an adjustable parameter, it cannot be *defined* in such a way as to make (12.1) correct.

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